ENGINEERING SCIENCE

Revised June 2020

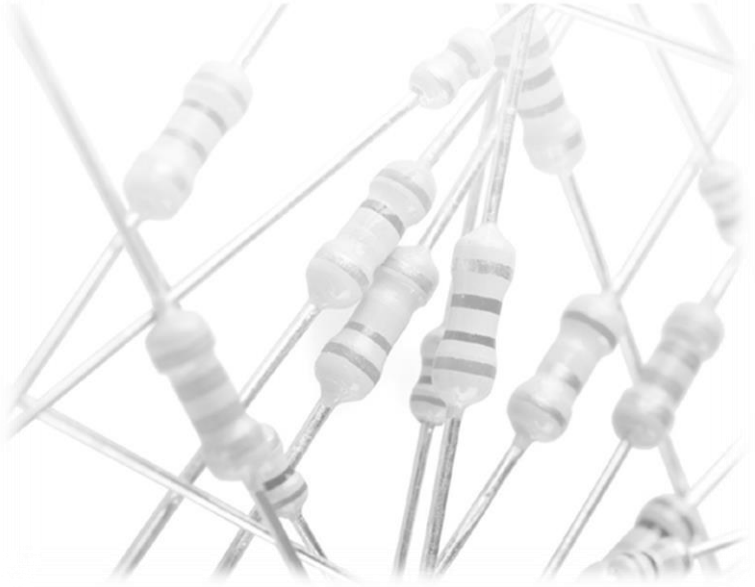
# Pupil booklet

## Electronics and control Analogue electronics

Topic 1: Electric circuits and components

Topic 2: Series circuits

Topic 3: Parallel circuits



Name: _____

Class: _____

Teacher: _____

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## *Learning Intentions*

- To know what an analogue component is.
- To know how to properly draw a circuit diagram using the proper symbols.
- To know about the basic concepts of current, voltage and resistance.
- To investigate simple series circuits.
- To investigate parallel circuits.
- To learn about symbols in circuit diagrams.
- To learn about LDR and thermistors.
- To learn about L.E.D.'s and protective resistors.

## *Success criteria*

Use traffic lighting to record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will sign off this page after speaking to you about your self-evaluation:

Red: I need to work on this topic further to understand it and ask for help.
Amber: I need a bit of help to fully understand this topic.
Green: I am confident in my understanding of this topic.

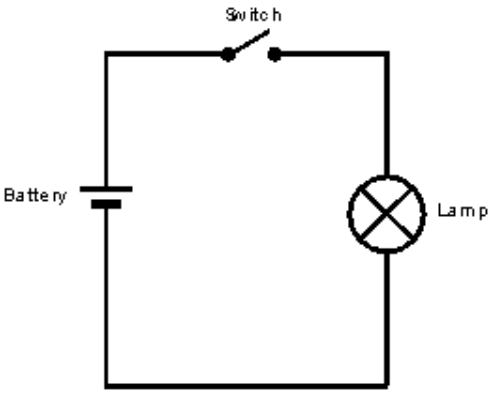
- ☐ I can identify the function and purpose within a circuit of: battery, switch, resistor, variable resistor, LDR, thermistor, LED, buzzer, diode, motor, lamp, ammeter and voltmeter.
- ☐ I can describe the function of a circuit in terms of input, process and output.
- ☐ I can perform calculations involving the relationship between voltage, current and resistance
- ☐ I can perform calculations involving resistors in series and parallel.
- ☐ I can perform calculations of voltage, current, and unknown values in a fixed-voltage divider.
- ☐ I can design a voltage divider to provide an input signal for a control circuit.
- ☐ I can interpret information given of characteristics for an LDR and an NTC thermistor.
- ☐ I can explain the function of a relay.
- ☐ I can explain the function of a protection diode in an electronic circuit.
- ☐ I can explain the switching function of a transistor.
- ☐ I can operate an electronic control circuit, which includes a variable voltage divider, transistor, relay and output transducer.

# Introduction to electricity

Electricity is one of the most important forms of energy available to man. It affects everyone's lives in many ways. If you take time to think about your everyday life you will realise that our lives are full of devices that depend upon electricity. These devices depend on the electrical circuits inside them to work. The circuits often change the electrical energy into other forms of energy such as heat, light and sound. As you have just learnt about in the unit 'system and control'. In this unit, you will learn how these circuits work and about the different components within them.

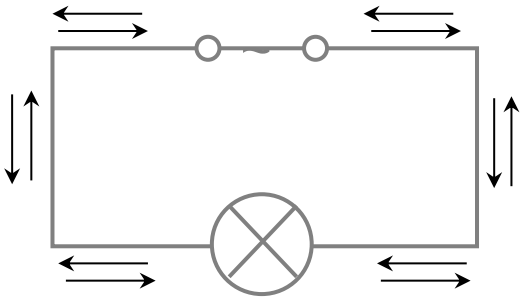
## Electric circuits

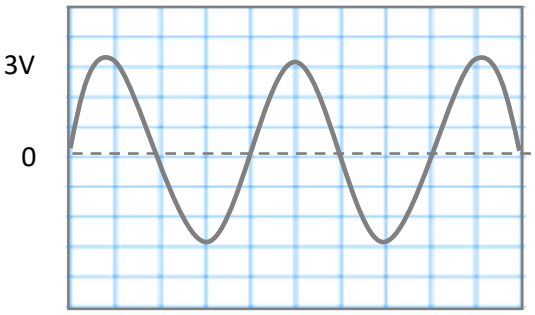
The diagram to the right is a basic series connected electric circuit. A series connected circuit is a closed loop or network, made up of electrical components such as batteries, bulbs, switches and wires. Copy this series circuit into your jotter.




## What powers a circuit?
## Alternating Current (a.c.) and Direct Current (d.c.)

The mains supply is a source of alternating current. The electrons move in one direction, then the other and back again. The current direction continuously alternates.

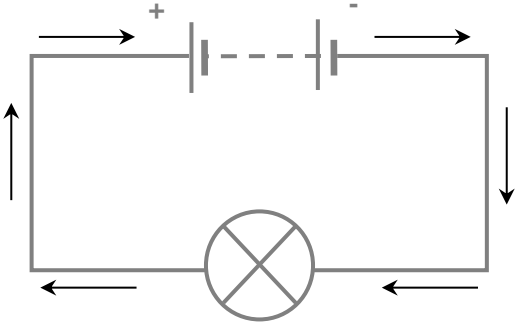


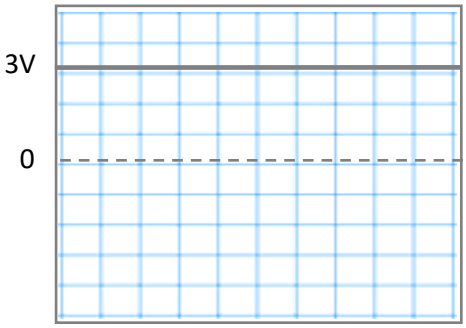


oscilloscope – a.c. signal

## Direct current (d.c.)

The voltage supplied by batteries or low-voltage supplies is direct current (d.c.). This is the normal type of supply to low-voltage circuits. Alternating current (a.c.) supplies are high-voltage, usually 230 volts. This is the normal supply in homes and schools. Many portable electric power tools work from 110 volts for safety. Batteries or cells are a common source of direct current. The negative charges (electrons) move from the negative side of the cell towards the positive side of the cell.

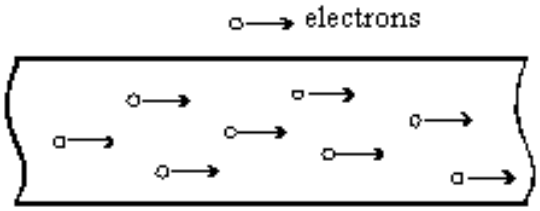


oscilloscope – d.c. signal

## *Electric current (I)*

The name given to the flow of negatively charged particles called electrons. Current is measured in *amperes*, usually referred to as 'amps' (A). Current is the rate of flow of electrical charges (called electrons) through a circuit.




## *Potential difference (Voltage) V*

A potential difference or voltage is a measure of the energy given to the charges. Specifically, it is the energy transferred by one coulomb of charge.

A 1.5V cell transfers 1.5 joules of energy to each coulomb of charge as it passes through the cell. This energy is then transformed into other forms by the components in the circuit.




In most circuits, a battery or Voltage Supply ($V_s$) is used to drive the electrons through the components.

The unit for potential difference is the volt (V).

Batteries and voltage supplies are the source of power behind all electrical circuits. Without a power source, electrical circuits would simply not work. The low-voltage supplies and batteries will normally supply between 3 and 12 volts.

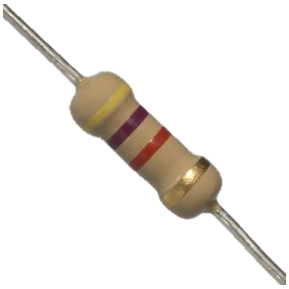
# Resistance (R)

All materials conduct electricity. The materials that conduct electricity well are called *conductors* and those that are poor conductors are called *insulators*. Metals are good conductors while rubber, wood and glass are good insulators.

A good conductor offers very little resistance to the flow of electrical current. In other words, it lets currents flow with very little voltage being applied. Resistance is therefore a measure of how much voltage is required to let a current flow. Resistance is measured in *ohms* (Ω).

## Resistors

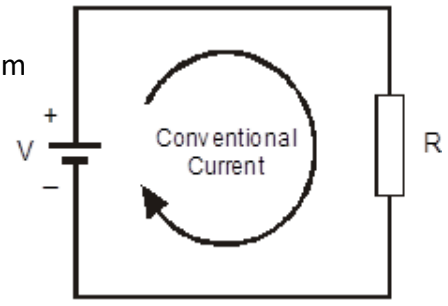
Resistors are basic components in electronic circuits. Resistors restrict or limit the amount of current flowing in circuits or parts of circuits. Resistors are cylindrical components and have coloured stripes. The stripes indicate the value of the resistors. The colours represent numerical values according to a special code. They also have connection wires sticking out of each end.



## Electron flow - conventional current

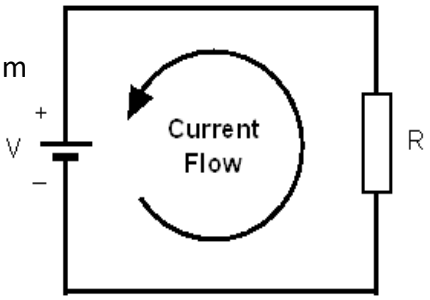
Scientists in the early nineteenth century decided the direction of conventional current flow. It seemed to them that current flowed from the positive side of power supplies to the negative side (diagram 1). It was not until the twentieth century that electrons were discovered and that the 'true direction' of current flow was proved of being the opposite (diagram 2).

Diagram




*(Conventional current flows from positive to negative)*

Diagram




*(Current flow actually flows from negative to positive)*

Electric current is the flow of electrons but often it is more useful to consider electric current to flow in the opposite direction. This is called *conventional current*. So although it is technically wrong, for convenience 'conventional current' will be used in the circuits and calculations throughout this work. One of the main reasons for maintaining this convention is that symbols and other data based on conventional current have become standard.

# Electric circuits

An electric circuit is a closed loop made up of electrical components such as batteries, bulbs and switches. Electric current is the name given to the flow of negatively charged particles, called electrons.

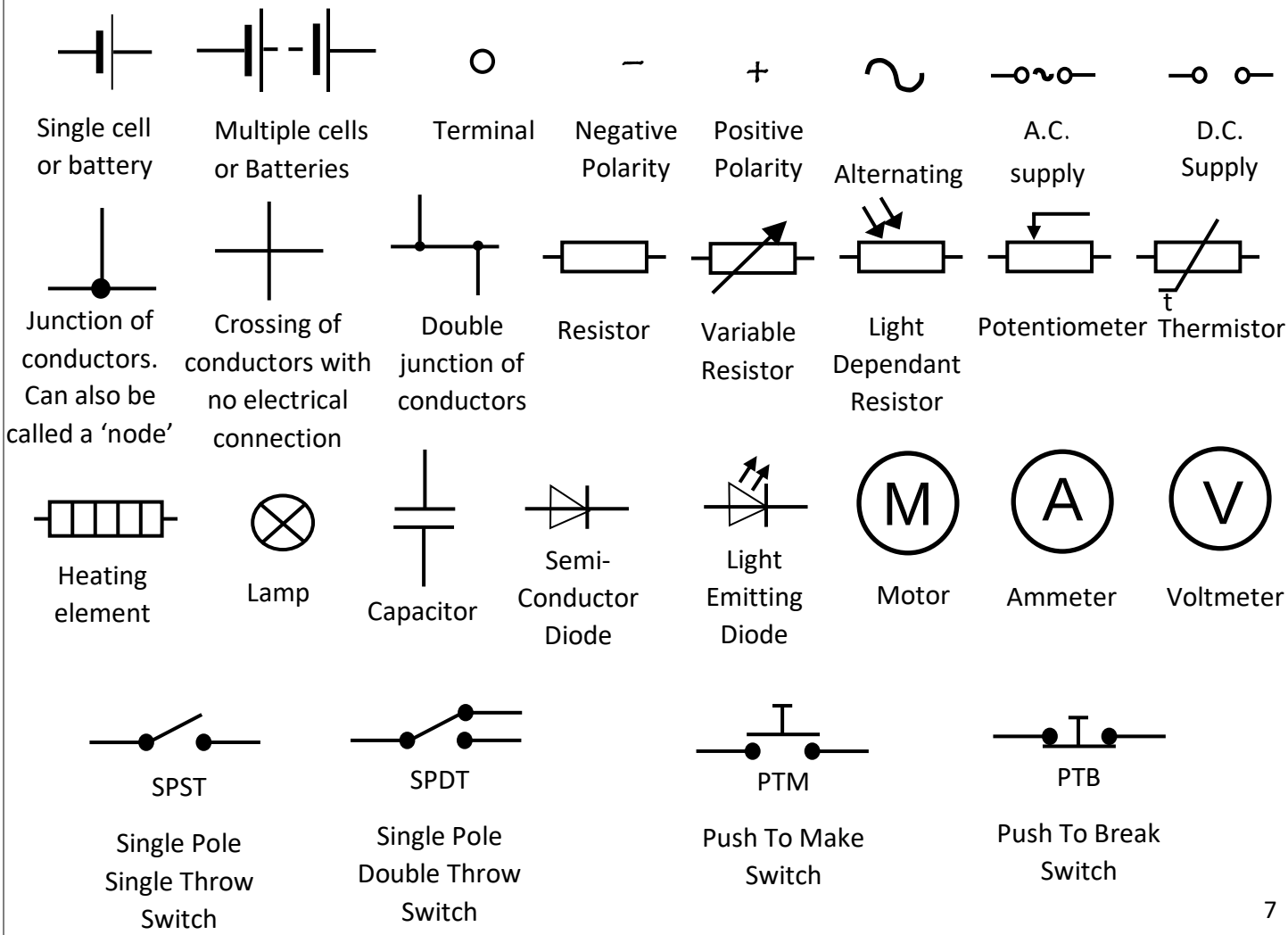
*'Conventional current'* is the direction in which an electric current is considered to flow in a circuit. By convention, the direction is that in which positive-charge carriers would flow – from the positive terminal of a cell to its negative terminal. It is opposite in direction to the flow of electrons. In circuit diagrams, the arrow shown on symbols for components such as diodes and transistors point in the direction of conventional current flow.

Current (I): is measured in amperes, usually referred to as amps (A). Current is the rate of flow of the electrons through the circuit.

Voltage (V): is used to drive/push the electrons through components in the circuit. Voltage is measured in volts (V).

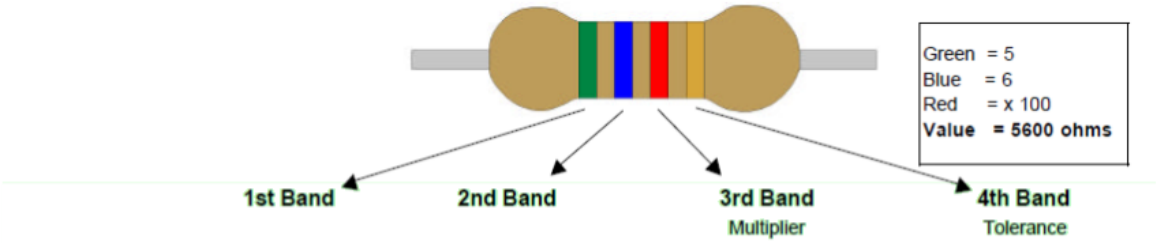
Resistance (R): is the measure of how much voltage is required to let current flow in a circuit. Resistance is measured in ohms (Ω).

## Electrical and electronic graphical symbols



## *Resistor colour codes*

Resistors are used for restrict or limit the amount of current flowing into a circuit and the extent to which they do this is measured in ohm's (Ω). In essence they act as a dam to current. Resistors are found in almost every electronic circuit. Resistors are marked with what is known as a resistor colour code. Each band of colour helps identify the value (in ohm's) and the tolerance (in percent). The colour code chart for resistors is shown below.

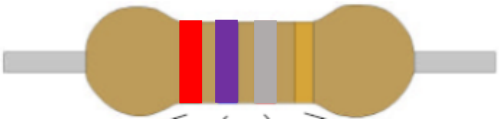



| | 1st Band | 2nd Band | 3rd Band (Multiplier) | 4th Band (Tolerance) |
|---|---|---|---|---|
| BLACK | 0 | 0 | x 1 | |
| BROWN | 1 | 1 | x 10 | ± 1% |
| RED | 2 | 2 | x 100 | ± 2% |
| ORANGE | 3 | 3 | x 1,000 | |
| YELLOW | 4 | 4 | x 10,000 | |
| GREEN | 5 | 5 | x 100,000 | |
| BLUE | 6 | 6 | x 1,000,000 | |
| VIOLET | 7 | 7 | | |
| GREY | 8 | 8 | | |
| WHITE | 9 | 9 | | |
| SILVER | | | x 0.01 | ± 10% |
| GOLD | | | x 0.1 | ± 5% |

**Example**

If the colours on the resistor are:

1st band - red
2nd band - violet
3rd band - brown
4th band - gold



Using the table above, the value of this resistor is 270 Ω and its tolerance is ±5 percent. This is worked out as '2' for the red first band, '7' for the violet second band and 'times 10' for the brown third band. For most purposes you can ignore the tolerance. In this example, the manufacturers guarantee that the resistor will not vary from the marked resistance by more than 5 percent.

## *Symbol for resistance*

The symbol for resistance of Ohms can sometime be shown in old circuit diagrams as a 'R'. 270 ohms can be expressed as either 270 Ω or 270 R. We would like you to use the symbol 'Ω' for the ohm's unit of measurement in calculations.

**1.** Collect the resistors from your teacher. Copy the table in your jotter and for each resistor write down the colour of each band and then calculate its value.

| Resistor | Band 1 | Band 2 | Band 3 | Tolerance band | Resistor value |
|---|---|---|---|---|---|
| 1 | | | | | |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |

**2.** Using the colour code chart, determine the colours of the first three bands of the following resistors.

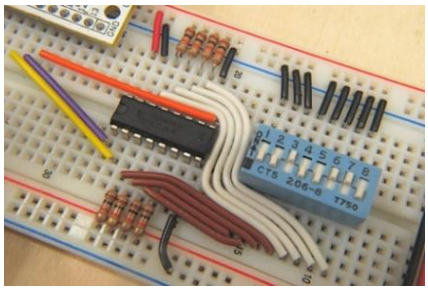
| No | Value | Resistor colour | Checked |
|---|---|---|---|
| 1 | 270Ω | Red, violet, brown | |
| 2 | 1.5kΩ | brown, green, red | |
| 3 | 33kΩ | orange, orange, orange | |
| 4 | 1.2MΩ | Brown red green | |
| 5 | 330Ω | | |
| 6 | 150Ω | | |
| 7 | 82Ω | | |
| 8 | 560Ω | | |
| 9 | 6.8kΩ | | |
| 10 | 750Ω | | |
| 11 | 390Ω | | |
| 12 | 2.1MΩ | | |
| 13 | 820Ω | | |
| 14 | 4700Ω | | |
| 15 | 9100Ω | | |

**3.** Using the colour-coding code, calculate the values of the following resistors.

| **No** | **Value** | **Band 1** | **Band 2** | **Band 3** |
|---|---|---|---|---|
| 1 | | red | red | red |
| 2 | | yellow | violet | black |
| 3 | | grey | red | red |
| 4 | | yellow | violet | orange |
| 5 | | red | red | orange |
| 6 | | orange | orange | orange |
| 7 | | green | blue | brown |
| 8 | | red | violet | black |
| 9 | | grey | red | brown |
| 10 | | brown | green | green |

| **No** | **Value** | **Band 1** | **Band 2** | **Band 3** |
|---|---|---|---|---|
| 11 | | brown | grey | yellow |
| 12 | | brown | black | yellow |
| 13 | | green | blue | orange |
| 14 | | brown | grey | black |
| 15 | | brown | grey | green |
| 16 | | blue | grey | orange |
| 17 | | orange | orange | yellow |
| 18 | | red | red | brown |
| 19 | | grey | red | black |
| 20 | | violet | brown | orange |

## *Breadboards*

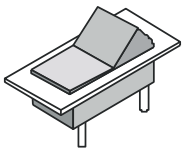


Breadboards are used to design practical circuits, without the need to solder or permanently join components. Breadboards are laid out like the one below. We can see that the top and bottom "rails" are connected horizontally. Also the middle rails are connected vertically.

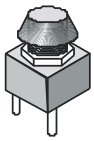
Your teacher may show you how to plan and build a breadboard circuit.

## *Switches*

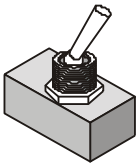
Switches are useful input devices (sometimes transducers) that have metal contacts inside them to allow current to pass when then they are touching. There are several ways in which the contacts in mechanical switches can be operated. The main types are - push-button, toggle, key, slide, magnetic (reed) and tilt. The switches below are a graphic representation of their physical appearance. The images do not represent how the switch would be indicated on a circuit drawing. These are 'digital' input devices as they can only be on or off.



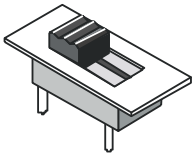
Rocker



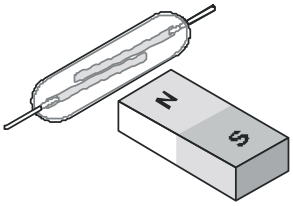
Push Button



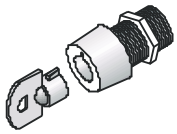
Toggle



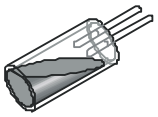
Slide




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Key

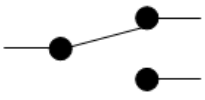


Tilt

Most of the switches you have seen so far are known as 'Single Pole Single Throw' but there are switches which allow switches that allow us to have more control over components. The image below is the correct graphic representation of a switch within a circuit diagram.



*SPST*
*Single Pole Single Throw*

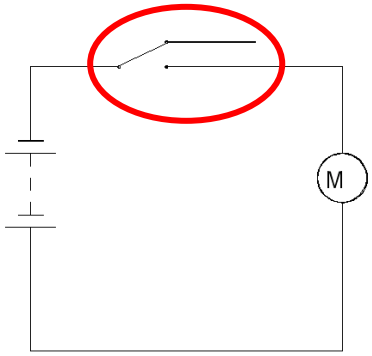


*SPDT*
*Single Pole Double Throw*

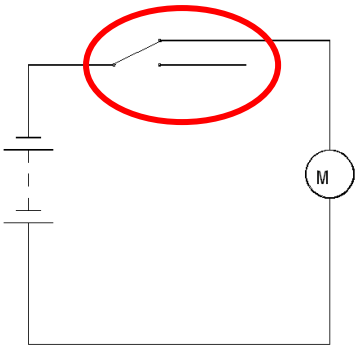
Micro switches which are Single Pole Double Throw can be wired up to produce a switch that is normally open (which means no current will flow until the switch is pressed and the circuit is a closed loop) or normally closed (which means the circuit is a closed loop and current is flowing to all components until the switch is pressed).

Different types of switch can also be wired up to suit their application. A switch with its contacts wired apart when it is not operating is also called Normally Open (NO). A switch with its contacts wired closed when it is not operating is called Normally Closed (NC). NO and NC circuits are illustrated on the next page.

Normally open




Normally closed




Push switches are a type of SPST (single-pole single-throw) switch.

They can be either push-to-make (PTM) switches, which are normally open (NO).



PTM

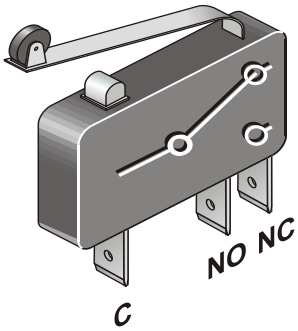
Push-to-break (PTB) switches, which are normally closed (NC).



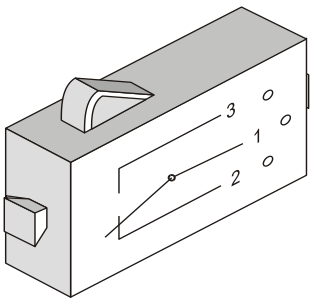
PTB

## *Microswitches*

Microswitches are small switches that are useful for detecting motion. They are especially good as sensors and limit switches. Typical systems that use microswitches are traffic barriers and lift systems.




The microswitch on the left has a roller fixed to a lever that detects movement and throws the switch. It has three terminals: common, normally open (NO) and normally closed (NC).




3 -- NO
1 -- C
2 -- NC

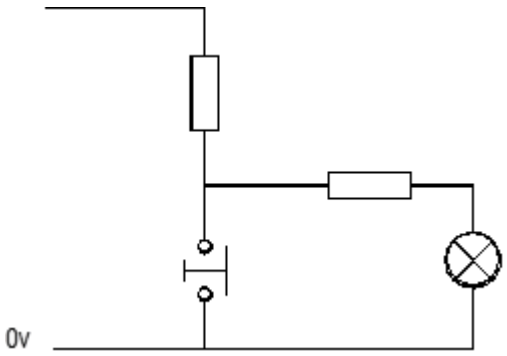
The microswitch on the right is commonly used in schools. Both of these microswitches are Single Pole Double Throw (SPDT) switches.

Like most microswitches, this one can be wired in three ways:

- C and NO: this is a normal on/off switch.
- C and NC: this allows current to flow when the switch is not operated.
- C, NC and NO: when wired like this it acts as a changeover switch.

## Digital switches

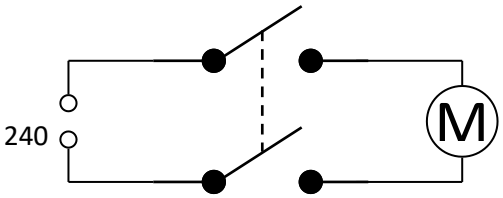
Simple PTM/PTB switches can be used in voltage divider circuits to give a digital signal (that is definitely on or off) to another part of a circuit. Current is a bit like water and will always take the path of least resistance.



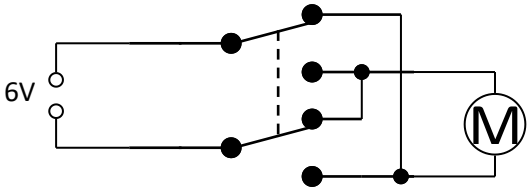

## DPST and DPDT switches

More complex switches exist to do more specialised tasks

The DPST switch (Double Pole Single Throw) is commonly used to isolate a piece of equipment or machinery from the power supply.

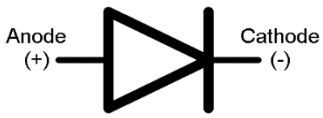



The DPDT switch (Double Pole Double Throw) is commonly used to allow a motor to switch directions.

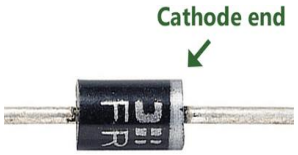



## Diodes

Diodes (symbol above) are devices that allow current to flow only in one direction. Current will flow through the diode only when the anode (positive side) is connected to the positive side of the circuit and the cathode (negative side) is connected to the negative side of the circuit.



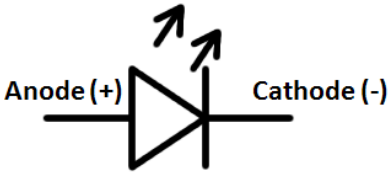

Graphic symbol
Direction of 'conventual' Current flow

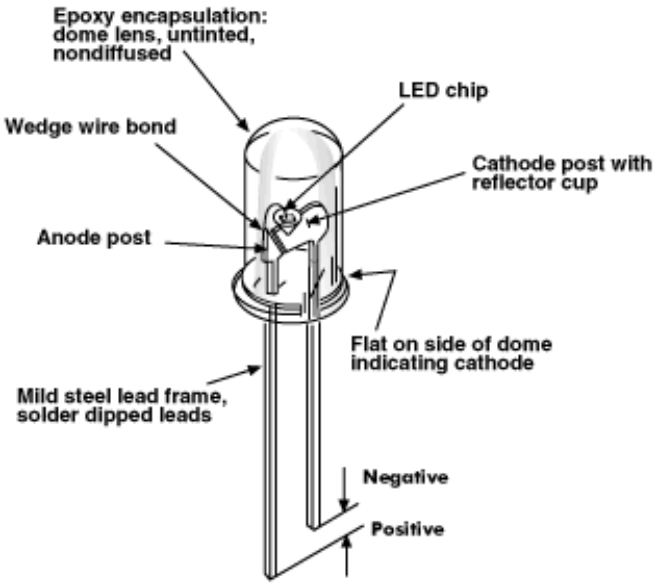



Actual image of a Diode component
Direction of 'conventual' Current flow

## Light-Emitting Diodes

A light-emitting diode (electrical symbol below) is a special diode that gives out light when current is flowing through it. LED's are used as indicators to tell when a circuit (or part or a circuit) is working. You can tell the cathode of an LED as it is the short leg and there is a 'flat' on the plastic casing. As with the normal diode, the current can only pass one way.

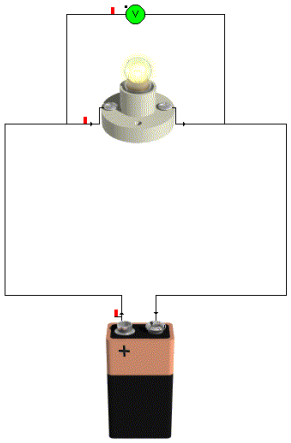


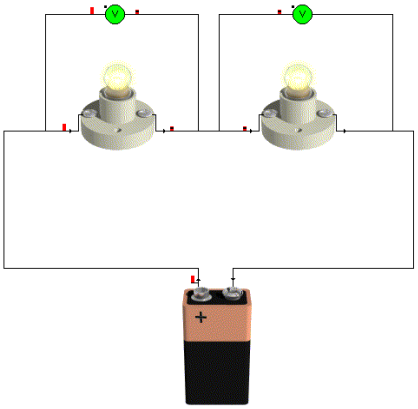
## *Series circuits*

**4.** Build the following circuit shown below in Yenka. In your jotter, draw each circuit diagram using the correct symbol for the circuit. Record the reading on the voltmeter for each circuit.

A)

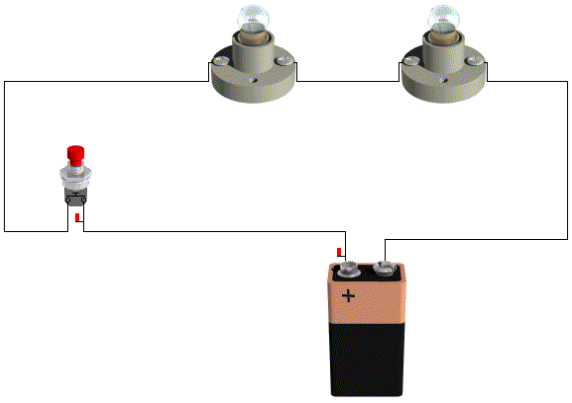


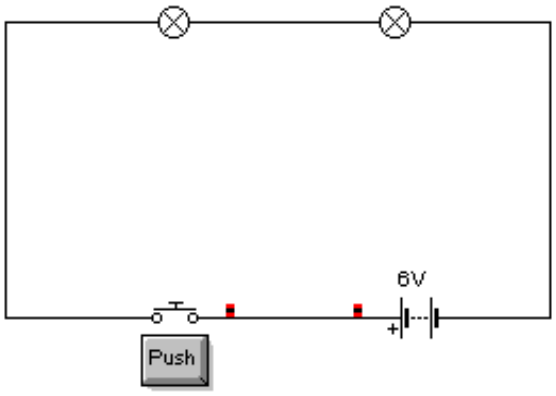
B)



## *Switches, variable resistors and motors*

**5.** The circuit below shows a simple series circuit which uses a push switch, one known as a "Single Pole Single Throw" (SPST). Build the following circuit shown below in Yenka. In your jotter, draw each circuit diagram using the electrical graphic symbols for the circuit.






Experiment the following tasks and record your findings:

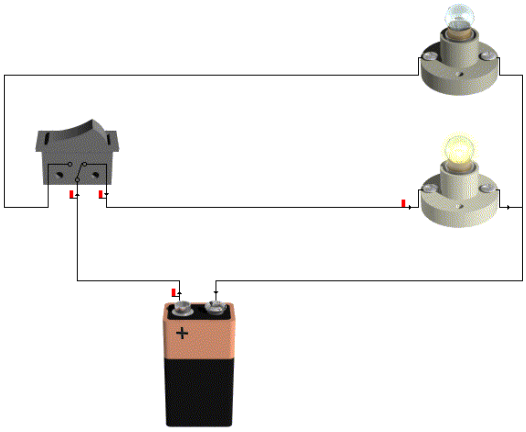
A) When the Switch is held down the bulbs will come on and when the button is released the bulbs will go off. Record the reading on the voltmeter in several sections of the circuit.

B) If the push switch is not being pressed the bulbs will go off. What type of switch should you use if you want the bulbs to stay on until the switch is pressed again?

C) Replace the push switch with a toggle switch (SPST) and test the circuit.

D) Draw the circuit diagram for the circuit you have just built in your jotter, using the correct circuit symbols.

**6.** The circuit below shows another type of rocker switch, a 'Single Pole Double Throw (SPDT)'. This one switch allows you to turn two bulbs but not at the same time. Build the following circuit shown below in Yenka. Draw the circuit diagram for the circuit you have just built, using the electrical graphic symbols.



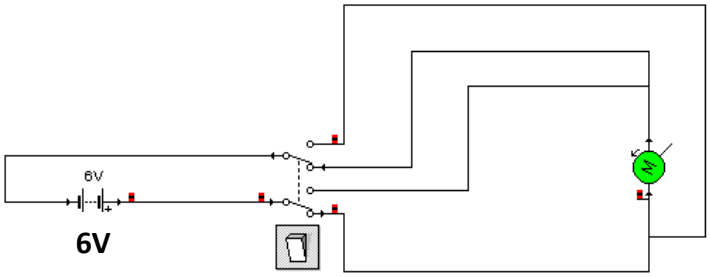
## *Real life contexts*

**7.** The circuit below uses a Double Pole Double Throw (DPDT) switch to control the direction of a motor.

A) Using Yenka, build and test the circuit. Draw the circuit in jotter.

B) Explain, using correct terminology how the circuit works.

C) Give two examples of where this could be used in a real life situation.



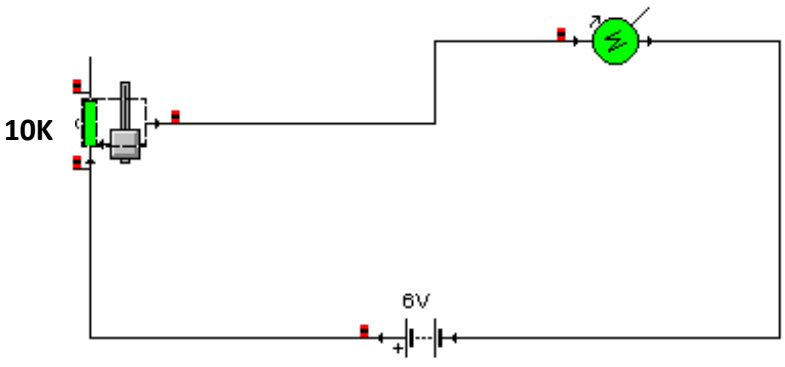

**8.** The circuit below shows a potentiometer being used to control the speed of a motor.

**Note:** A potentiometer acts exactly like a variable resistor. They do exactly the same job

A) Using Yenka, build and test the circuit. Draw the circuit in jotter.

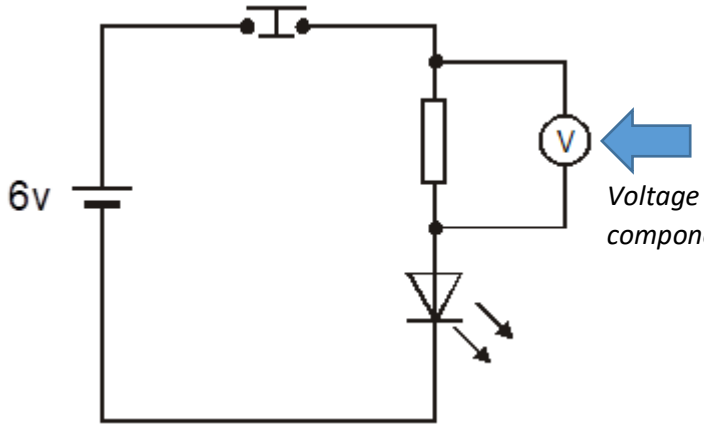
B) Explain, using correct terminology how the circuit works.

C) Give two examples of where this could be used in a real life situation.



## *Measuring voltage and current in a series circuit*

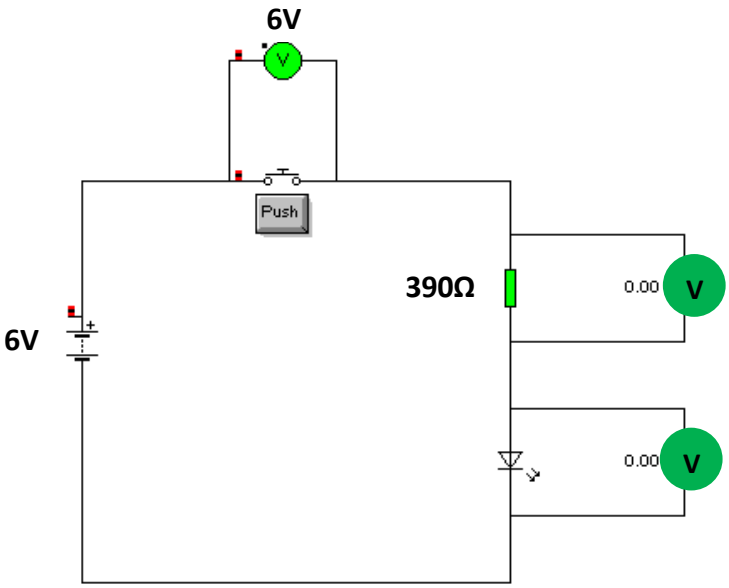
To measure the voltage used up by a component or the 'voltage dropped', we connect the voltmeter in parallel with the component, as shown below. As the current in a series circuit is the same all the way through the circuit we only need to take a reading from one place in the circuit. To do this we must connect it up in series with the other components.




*Voltage measured in parallel to the component*

Remember: Voltage is measured <u>across</u> components or parts of circuits (parallel) and Current is measured through the circuit (series).

**9.** Using Yenka, measure the voltage across all three components in the LED circuit.




The voltage across each component is known as the *voltage drop* across the component. This is the amount of voltage 'used up' or 'dropped' by each. The total voltage dropped in the circuit should equal the total supply voltage as stated in Kirchoff's second law. This second law is explained next.

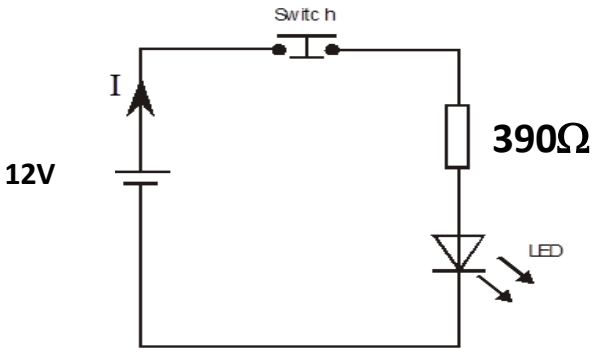
In your jotter, draw the circuit diagram using the electrical graphic symbols for the circuit and record your results.

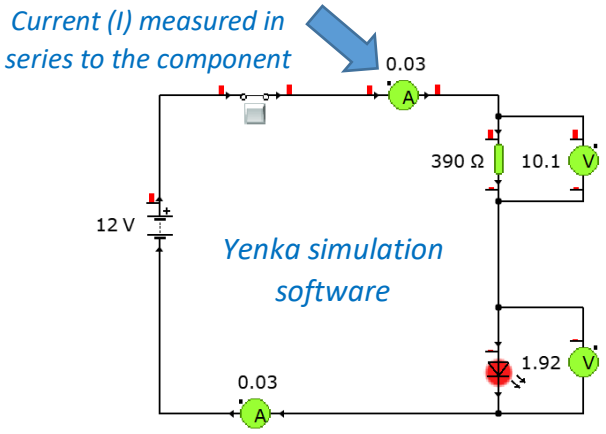
# *Measuring direct current*

Current is measured through components or parts of circuits, as shown in the circuit diagram below. Note that it is necessary to 'break' the circuit and connect the meter in series with the components to measure current unlike voltage.

**10.**

a) In your jotter, draw the series circuit (shown below) using the electrical circuit symbols and the table. Record the current measured in Amps (A) within the circuit and the voltage across the 1 LED.




| Circuit | Mains voltage (V) | Resistor $R_1$ value (Ω) | Voltage across LED 1 (V) | Resistor $R_2$ value (Ω) | Voltage across LED 2 (V) | Voltage across LED 3 (V) | Overall circuit resistance (Ω) | Circuit current ammeter value Amps (A) |
|---|---|---|---|---|---|---|---|---|
| 1 | 12 | 390 | | X | X | X | | 0.03 A (1 s.f.) (as shown in Yenka above) |

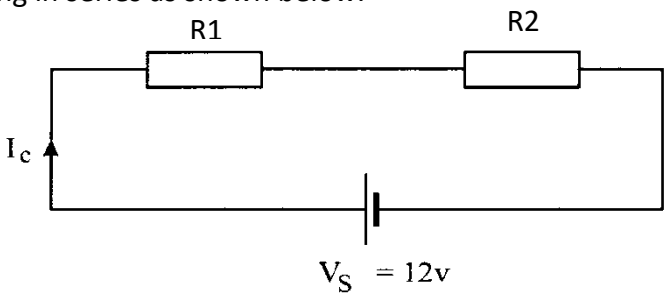
b) To increase the precision of ammeter measurement, 'right click' on the ammeter and select properties. Change the Autorange to 'off' and Measure RMS to 'on' (green tick).

c) Using Yenka and the same circuit above, experiment the following resistors and components in SERIES on a prototype circuit board and measure the overall resistance, circuit current and voltage over each component. In your jotter, copy the table below and draw each circuit diagram using the correct symbol for the circuit and record the results over each resistor. Note: circuits 1 to 4, only have 1 LED. Circuit 5 has 2 LED's and circuit 6 has 3 LED's. Extension: measure the voltage across each component.

| Circuit | Mains voltage (V) | Resistor $R_1$ value (Ω) | Voltage across LED 1 (V) | Resistor $R_2$ value (Ω) | Voltage across LED 2 (V) | Voltage across LED 3 (V) | Overall circuit resistance (Ω) | Circuit current ammeter value (A or mA) |
|---|---|---|---|---|---|---|---|---|
| 1 | 12 | 390 | | X | X | X | | mA (3 s.f.) |
| 2 | 12 | 350 | | X | X | X | | mA (3 s.f.) |
| 3 | 12 | 300 | | 400 | X | X | | mA (3 s.f.) |
| 4 | 12 | 350 | | 350 | X | X | | mA (3 s.f.) |
| 5 | 12 | 350 | | 350 | | X | | mA (3 s.f.) |
| 6 | 12 | 350 | | 350 | | | | mA (3 s.f.) |

## *Resistors connected in series circuit*

Resistors come in standard sizes. When resistors are connected one after each other to obtain a specific value. This is called connecting in series as shown below.




To calculate the total overall resistance of a number of resistors connected in this way you add up the individual resistances. Two resistors in series gives the following formula:

$$R_{Total} = R_1 + R_2$$

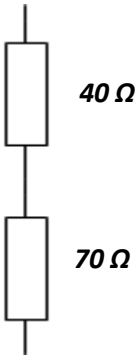
Example: To calculate the total resistance for <u>three</u> resistors in series:

$$R_{Total} = R_1 + R_2 + R_3$$
$$= 100\ \Omega + 82\ \Omega + 1\ \Omega$$
$$= \underline{183\ \Omega}$$

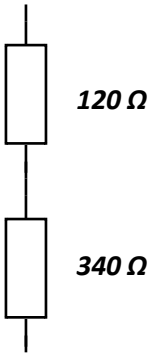
*Note: Resistors in series $R_t = R_1 + R_2 + R_3 + \ldots$ equation provided in data booklet.*

**11.** Copy the diagrams into your jotter and calculate the combined resistance of the following resistors in series.

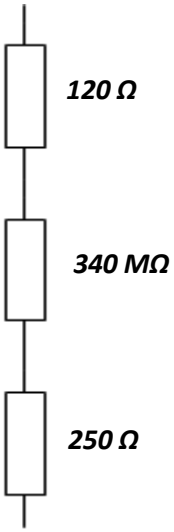
A)



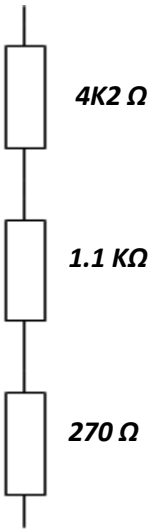

B)




C)



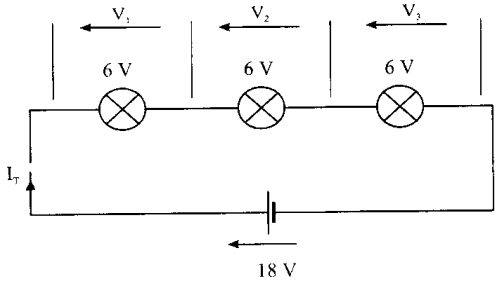

D)




## *Kirchoff's 2nd law*

Kirchhoff's Voltage Law (KVL) is Kirchhoff's second law that deals with the conservation of energy around a closed circuit path. His voltage law states that for a closed loop series path the algebraic sum of all the voltages around any closed loop in a circuit is equal to zero. This is because a circuit loop is a closed conducting path so no energy is lost.

An example of a 3 bulb series circuit is shown.



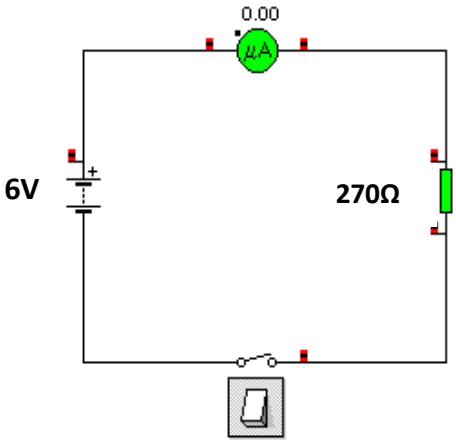

Kirchoff's 2nd law series circuit: (provided in data booklet)

$$\text{Voltage Supply } (V_t) = V_1 + V_2 + V_3$$

## *Ohm's law*

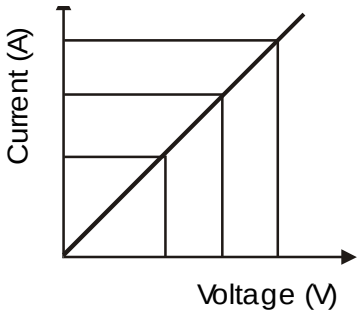
Ohm's law states that the current in a series circuit is proportional to the voltage drop across a resistor. You have already seen that applying a voltage to a circuit results in a current flowing through the circuit.

In the Yenka circuit opposite, if you double the voltage across a component, the current flow through that component will also double which can be dangerous. Thus we can say that the current is 'equally proportional' to the voltage drop across a resistor as demonstrated in the current V voltage graph below. This rule is known as *Ohm's law*. The rule applies to all metals, provided that their temperature does not change.




This relationship gives rise to the Ohm's law formula:

$$R = \frac{V}{I}$$



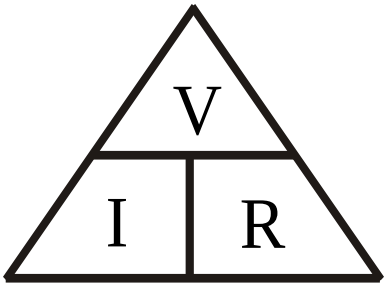

which is more easily remembered as:

**V = IR**

**Voltage = Current x Resistance**

*Note: Ohm's law V = IR equation provided in data booklet.*

We can use the triangle trick to help transpose this formula. Cover up the quantity that you are trying to find and the other two will be in the form that is needed.




## *Ohm's law in practice*

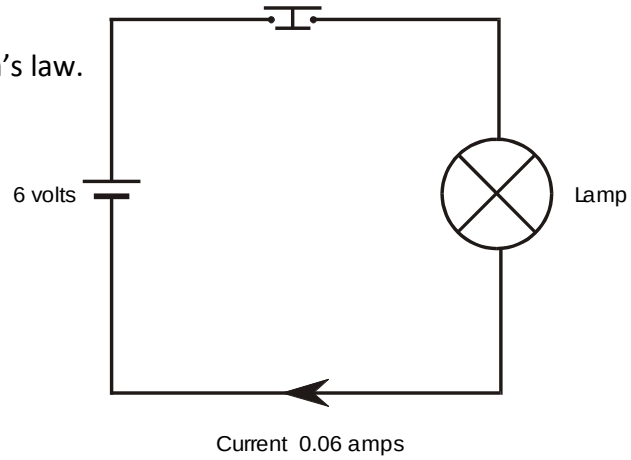
This exercise a simple series circuit is used.

The task is to calculate the resistance of the lamp using Ohm's law.

$$R = \frac{V}{I}$$

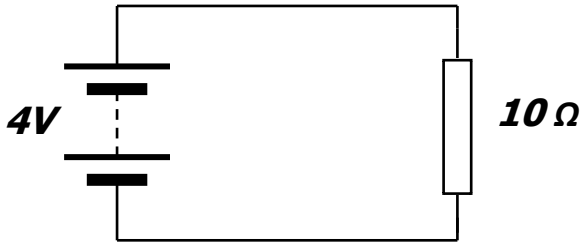
$$R = \frac{6}{0.06}$$

$$\therefore R = 100\Omega$$

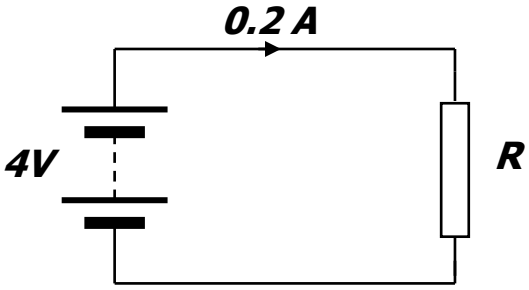


**12.** In your jotter, draw the following circuits and calculate the missing quantity. You can verify your answer with Yenka.

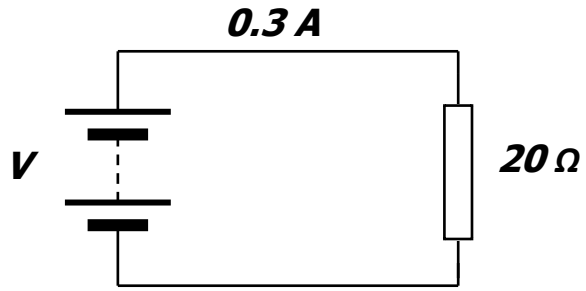
a) Calculate the current



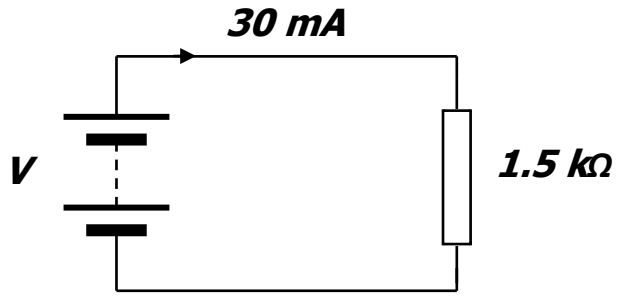

b) Calculate the resistance of the resistor R




c) Calculate the battery voltage

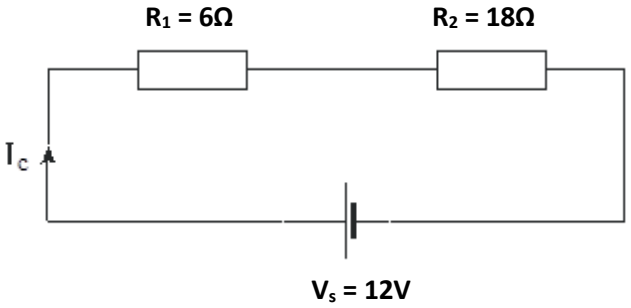



d) Calculate the battery voltage




Don't forget: mA means milliamps (1000th of an amp). All unit values (KΩ and MA) can quantities be found in your data booklet and answers to two significant figures.

## Worked example: series circuit




1. In your jotter, draw the series circuit shown and calculate:

a) the total resistance ($R_T$)
b) the circuit current ($I_C$)
c) the potential difference across both resistors ($V_1$ and $V_2$)

a) the total resistance ($R_T$)

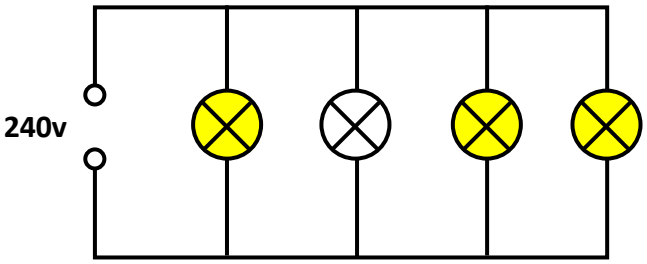
b) the circuit current ($I_C$)

c) the potential difference across both resistors ($V_1$ and $V_2$)

We can use Kirchoff's second law to check the answers calculated for the potential difference across the resistors:

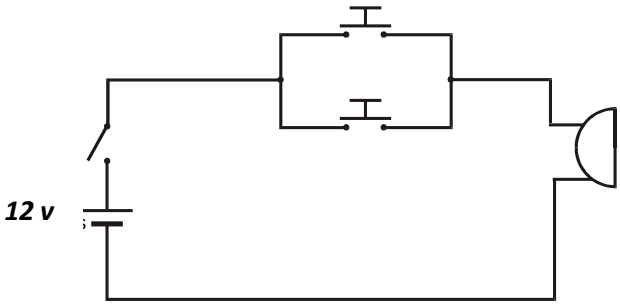
## Parallel circuits

Parallel circuits act differently to series circuits. In this case, voltage is equal throughout each of the parallel branches. Current however will be different in each branch. If one bulb fails, the current can still travel through the other branches, so the rest of the circuit will still work. Each branch receives the supply voltage, which means that you can run a number of devices from one supply. A good example is a set of Christmas tree lights where all the bulbs require a 240V supply.





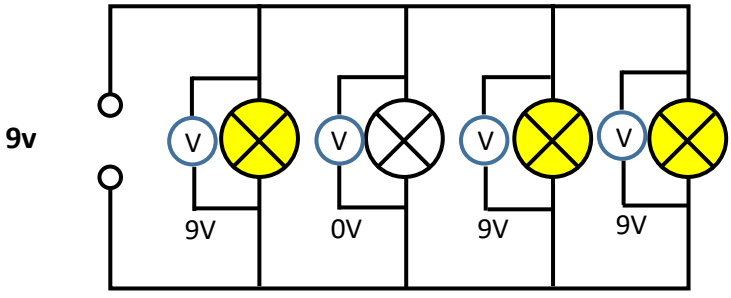

Parallel circuits can be arranged in many ways, but are normally set out so that you can easily see the parallel 'branches'. A simple parallel car-alarm circuit is shown below with the switches wired up in parallel. The two switches represent the sensor switches connected to the doors.




## Measuring voltage and current in a parallel circuit

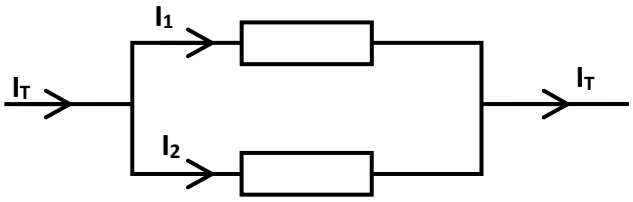
### Voltage in a parallel circuit

Within a parallel circuit, each branch receives the supply voltage. This means that you are able to run a number of devices from one supply. And they will receive the same input voltage.



## *Current in parallel circuits*

As we know, each branch of a parallel circuit receives the total supply voltage - but this is not the same for current (I). When the current reaches a junction, it splits up, with some current going along one branch and the rest goes along the other branch, as shown in the diagram below.

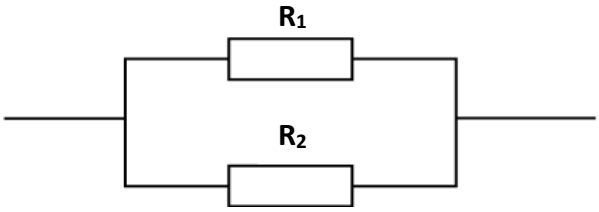



Kirchoff's 1st law
$I_T = I_1 + I_2 + I_3 + \ldots$
Provided in data booklet

## *Resistance in parallel circuits*

If connecting resistors in series circuit does not achieve the desired resistance, connection in parallel can also be used.

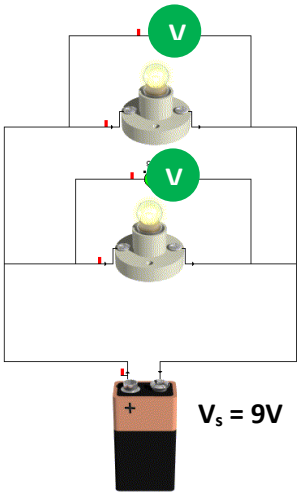
To calculate resistance in a parallel circuit:




$$R_T = \frac{R_1 \times R_2}{(R_1 + R_2)} \quad \text{or} \quad \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Note: The first $R_T$ equation can only be used if there are just <u>two</u> resistors connected in parallel. The second equation can be used for <u>any</u> number of resistors in a circuit. (Both resistors in parallel equations provided in data booklet.)

**13.** In your jotter, draw the circuit opposite using the electrical graphic symbols. Record the voltage results over each bulb. The supply voltage is 9V. You can verify your answer with Yenka.




## *Power in electrical circuits*

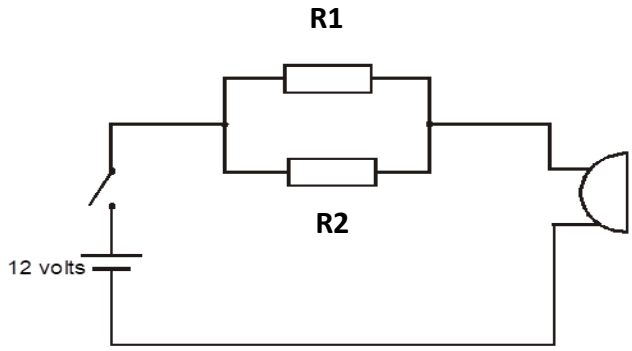
Electrical power (P) is measured in watts (W). Electrical power can be converted into other forms of power using electrical circuits. A good example is an electric fire, which has a heating element. The power used in overcoming the electrical resistance can be converted into heat.

The power in an electric circuit depends both on the amount of current (I) flowing and the voltage (V) applied. To calculate the Power in a circuit you can use the following rule:

$$P = IV$$

**Note:** Power calculations will be in the Energy and Efficiency unit.

## *3 worked examples: Resistors in parallel*




Example 1. In your jotter, draw the circuit shown and calculate:

a) The total resistance of the parallel part of the circuit.
b) The total resistance in the circuit. The resistance values are $R_1$ = 270Ω, $R_2$ = 100Ω and for the buzzer 240Ω.

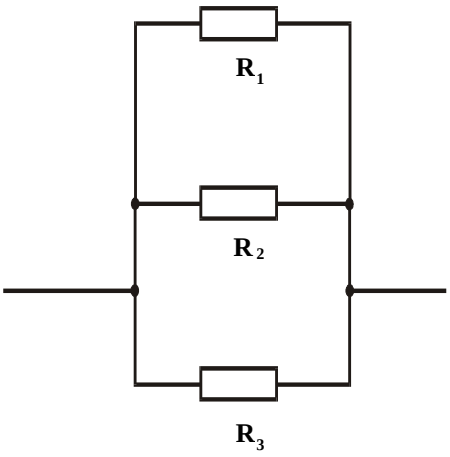
a) The total resistance of the parallel part of the circuit. (there are two ways of calculating two resistor total when in parallel.)

a) POWERPOINT PRESENTATION WORKED ANSWER

Example 2. In your jotter, draw the circuit shown and calculate:

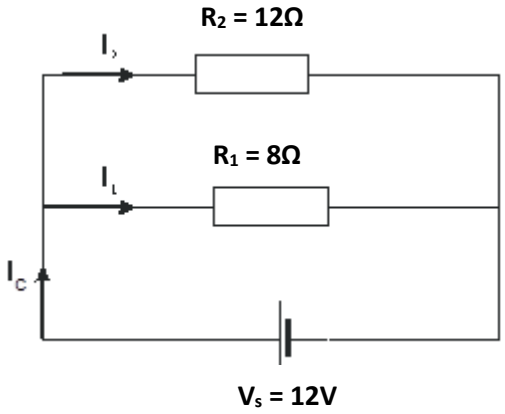
a) The total resistance in the circuit. The resistance values are $R_1$ = 220 Ω, $R_2$ = 100 Ω and $R_3$ = 330 Ω.

POWERPOINT PRESENTATION WORKED ANSWER



Example 3. In your jotter, draw the circuit shown and calculate:

a) The total resistance ($R_T$).
b) The circuit current ($I_C$).
c) The current in each resistor ($I_1$ and $I_2$).




a) The total resistance ($R_T$). (using the alternative 'resistors in parallel' equation when compared to worked example 1). This equation can ONLY be used with TWO resistors in parallel.

POWERPOINT PRESENTATION WORKED ANSWER

b) The circuit current ($I_C$).

POWERPOINT PRESENTATION WORKED ANSWER

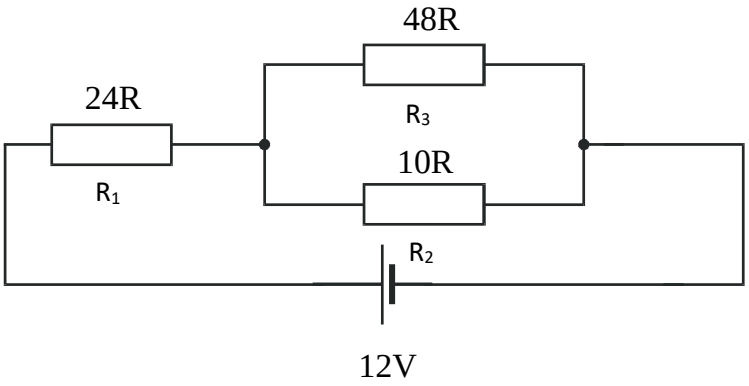
c) The current in each resistor ($I_1$ and $I_2$ then prove the total current $I_c$).

POWERPOINT PRESENTATION WORKED ANSWER

## Worked example: combined series and parallel circuits

1. In your jotter, draw combined series and parallel circuit shown and calculate:

a) The total circuit resistance ($R_T$).
b) The circuit current ($I_C$).
c) The voltage drop across resistor $R_1$ ($V_{R1}$).
d) The voltage drop across resistor $R_2$.
e) The current through resistor $R_2$ ($I_2$).




a) ($R_P$) of $R_2$ and $R_3$. It is possible to use the special case formula for two resistors in parallel:

($R_p$ = Resistors in parallel)

POWERPOINT PRESENTATION WORKED ANSWER

b) Calculate the current in the whole circuit ($I_c$):

POWERPOINT PRESENTATION WORKED ANSWER

c) Calculate the voltage drop across $R_1$. Use the current through the whole circuit ($I_c$) and the resistance of $R_1$:

POWERPOINT PRESENTATION WORKED ANSWER

d) Calculate the Voltage drop across the resistors in parallel. By using Kirchoff's 2nd law we know that the voltage drop across the parallel arrangement must be:

POWERPOINT PRESENTATION WORKED ANSWER

e) By using Kirchoff's first law we know that the circuit current $I_C$ will 'split' or divide between the two resistors $R_2$ and $R_3$. The current through $R_2$ is found by using the resistance of $R_2$ and the voltage drop across $R_2$.

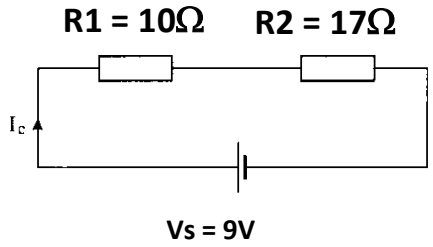
POWERPOINT PRESENTATION WORKED ANSWER

**14.** In your jotter, draw the following series circuits and calculate:

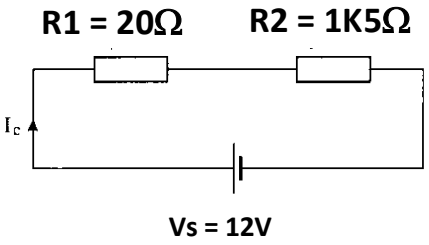
a) The total resistance ($R_T$).
b) The circuit current ($I_C$). (I = V/R)
c) The voltage across each resistor. (V = IR)

You can verify your answers with Yenka.

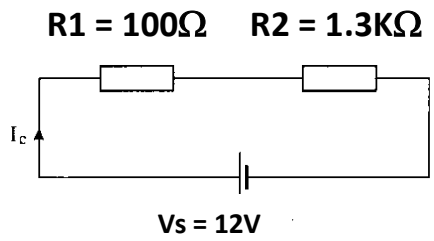
a)



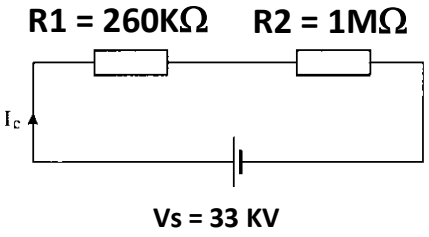

b)




c)




d)

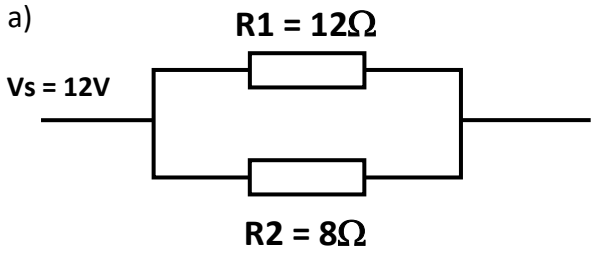



**15.** In your jotter, draw the following parallel circuits and calculate:

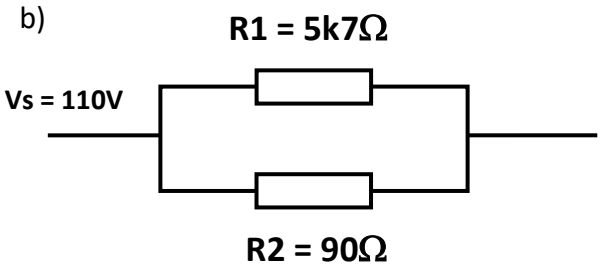
a) The total resistance ($R_T$).
b) The circuit current ($I_C$). ($I_c = V_s/R_T$)
c) The current across each resistor ($I_1$ and $I_2$ and so on).

You can verify your answers with Yenka.

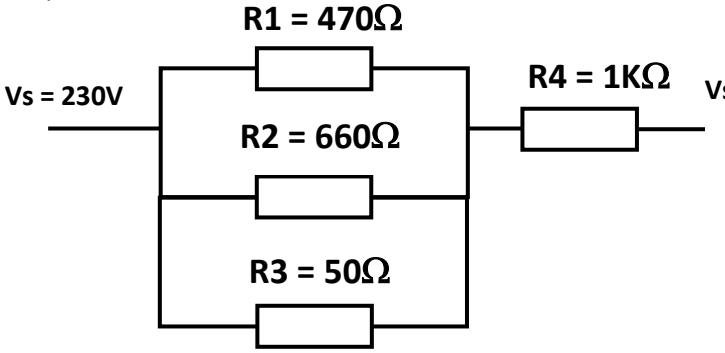
a)



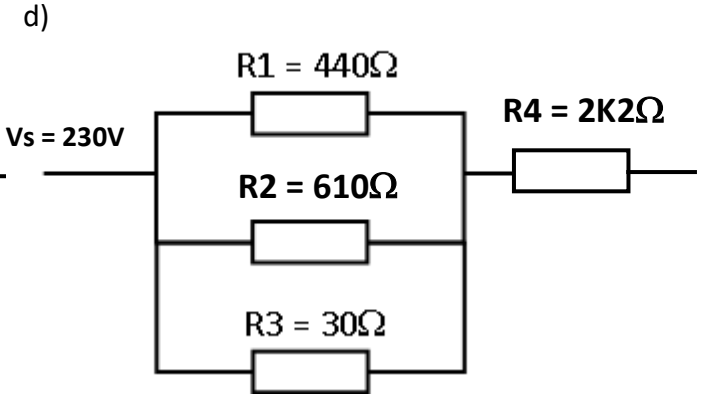

b)




c)




d)



**16.** Answer the following questions in your jotter. You will need to draw a circuit diagram.

**1.** Two resistors (15kΩ, 25kΩ) are connected in parallel and then connected to a 6v supply voltage. Calculate;

- **a)** The total resistance.
- **b)** The circuit current.

**2.** Three resistors (2kΩ, 5kΩ, and 10kΩ) are connected in parallel and then connected to a 10v supply voltage. Calculate;

- **a)** The total resistance.
- **b)** The circuit current.
- **c)** Current through the 2kΩ resistor.

**3.** A burglar alarm has 2 buzzers connected in parallel, which have a resistance of 200Ω and 300Ω and are connected to an 8v supply voltage. Calculate;

- **a)** The circuit resistance.
- **b)** The circuit current.
- **c)** The current through the 200Ω buzzer.

**4.** A student needs a 50Ω resistance to complete her electronics project. Unfortunately she only has two 100Ω available. Draw a solution to her problem and explain your answer by providing proof (calculation) that it is correct.

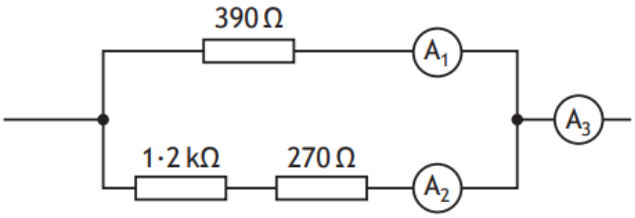
**5.** A circuit with a voltage supply of 6v has a 30Ω and a 20Ω resistor connected in parallel and then they are connected in series to a 10Ω resistor. Draw the circuit and calculate;

- **a)** The total resistance of the circuit.
- **b)** The circuit current.
- **c)** The voltage dropped over the 10Ω resistor.
- **d)** The current through the 30Ω resistor.

**6.** **N5 past paper 2019 Q8.** An interactive speaker used in a 'smart home' is shown.



Part of the circuit used in the speaker is shown below.




a) Calculate the total resistance of this circuit — 3 marks

b) The reading on ammeter A1 is 0.031 A.

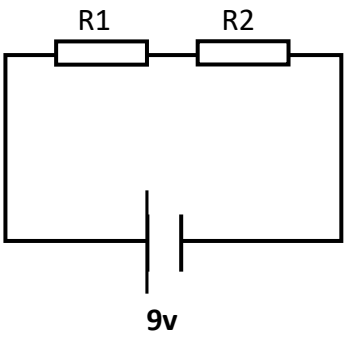
(i) Calculate the voltage across the 390 Ω resistor. — 3 marks

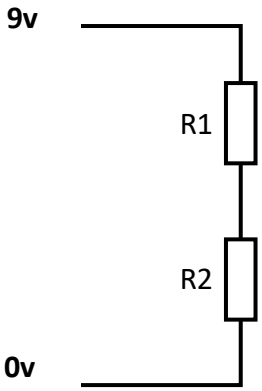
(ii) Calculate the current A2. — 3 marks

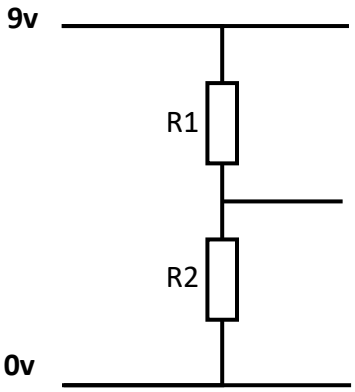
(iii) Calculate the current A3. — 1 mark

## *Input transducers*

Input transducers are devices that convert a change in physical conditions (for example a thermistor, LDR, or variable resistor) into a change in resistance and/or voltage. This is then processed by a voltage divider circuit. Below are some examples of how a voltage divider can be drawn.

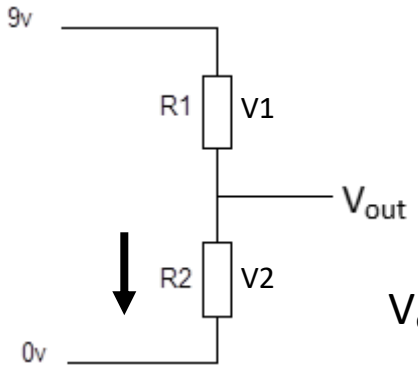



## *Voltage divider*

A voltage divider is a simple circuit which turns a large voltage into a smaller one. The circuit below is a typical voltage divider which by using just two resistors connected in series and an input voltage, you can create an output voltage that is a fraction of the input. If the value of R1 is changed the voltage across it will change, as will the voltage of R2.




$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

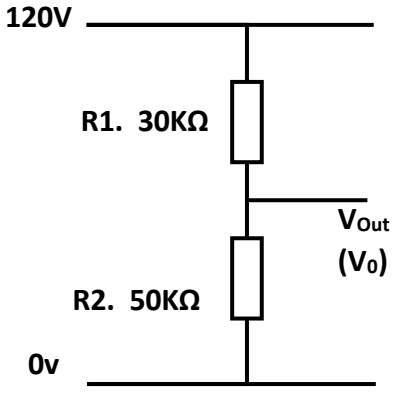
*Note: The Voltage divider equation on the left is the only one provided in the data booklet to calculate unknown quantities.*

$$V_{out} = V_{supply} \times \left[ \frac{R_2}{R_1 + R_2} \right]$$

*Note: Vout equation is NOT provided in the data booklet.*

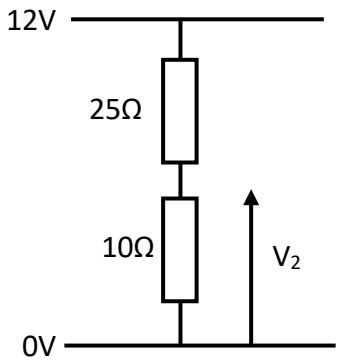
The voltage over R1 added to the voltage over R2 must be equal to the supply voltage.

## Worked example. Calculate $V_0$:

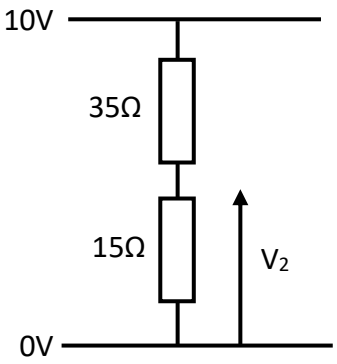


**17.** In your jotter, draw the following circuits and calculate the missing value. You can verify your answers with Yenka.

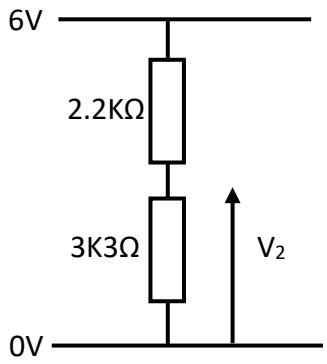
**A.**



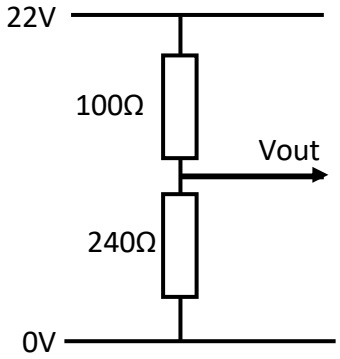

**B.**



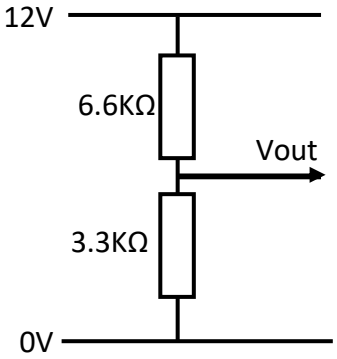

**C.**



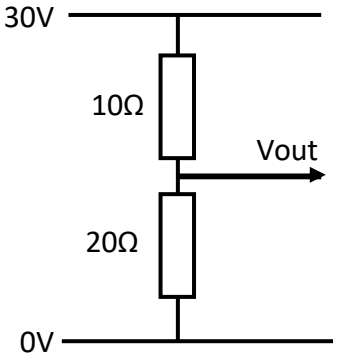

**D.**



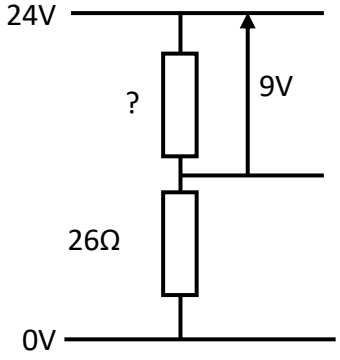

**E.**



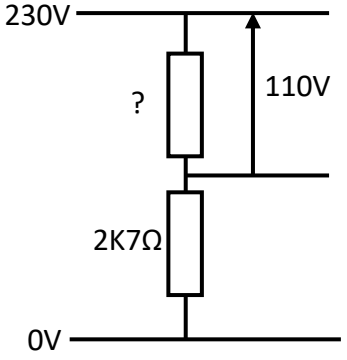

**F.**



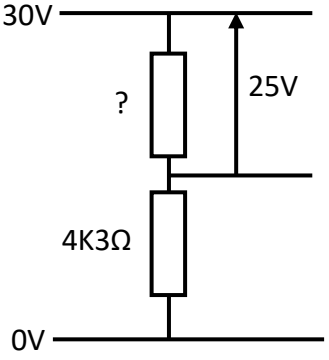

**G.**



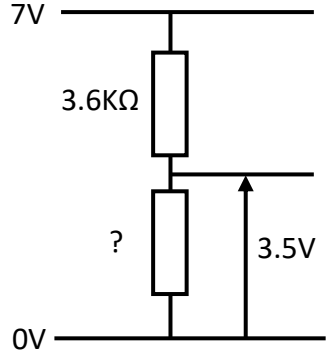

**H.**



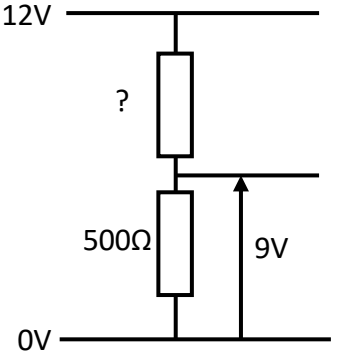

**I.**



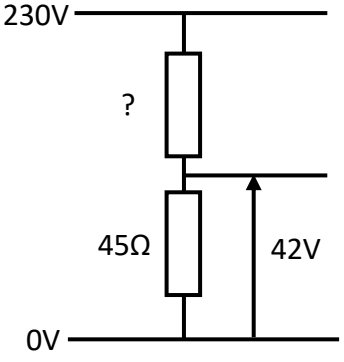

**J.**




**K.**




**L.**



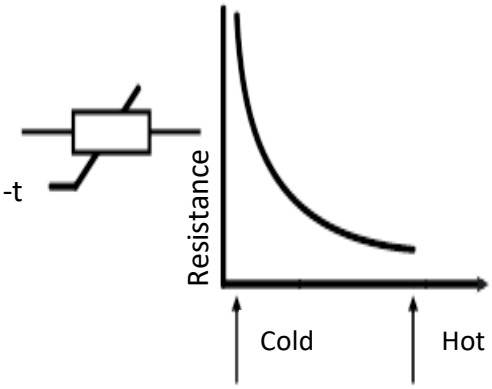
# More complex circuits

Now that we have learnt the basics, we need to apply these concepts to more complex circuits. The two most common analogue input transducers are the thermistor and the light dependent resistor (LDR).

## Thermistor

Thermistor – a thermistor is a device whose resistance varies with temperature. It is a temperature dependent resistor. There are 2 main types:

1. Negative Temperature Coefficient (-t or NTC). As the temperature increases, the resistance will decrease. The voltage across the thermistor will also decrease (as V = IR). These are the most commonly used.



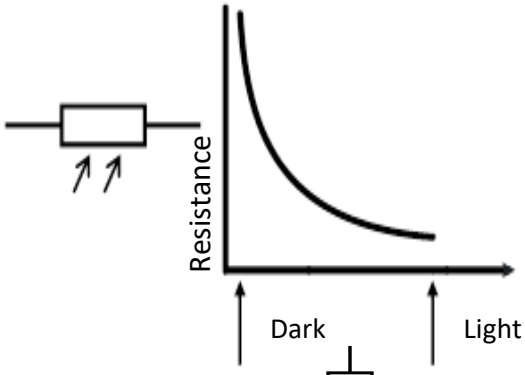

**T**emperature **U**p, the **R**esistance does **D**own: 'TURD'

2. Positive Temperature Coefficient (+t or PTC). As the resistance increases, the temperature also increases.

## Light Dependent Resistor (LDR)

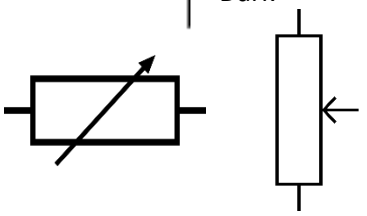
This is sometimes referred to as an LDR.

As the **L**ight goes **U**p, the **R**esistance goes **D**own: 'LURD'




## Variable resistor / Potentiometer

These can be used as a voltage or current control device. These are often used in a voltage divider circuit to adjust the sensitivity of the circuit.



Variable resistor. Potentiometer.

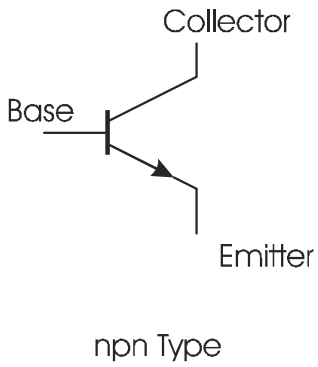
## Bipolar transistor

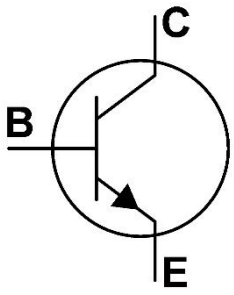
The transistor is a semiconductor device and acts as a switch. This means that it is sometimes a good conductor of electricity and sometimes a poor one. A transistor is made up of three layers of semiconductor materials that are either 'n type' or 'p type'. We will deal only with the 'npn' type for convenience. The transistor must be connected into circuits correctly. The arrowhead on the emitter indicates the direction of 'conventional' current flow. Conventional current was explained at the start of this unit.

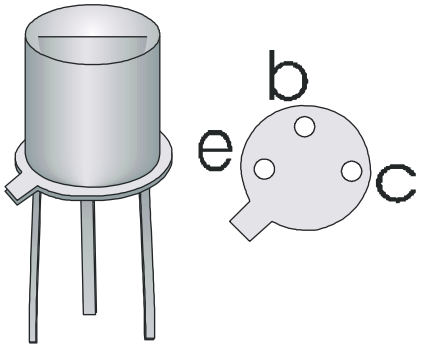
When a positive voltage of about 0.6 Volts is applied across the base and emitter, the resistance between the collector and the emitter of the transistor drops from very high to very low. In other words, the transistor changes from being a very poor to a very good conductor.

This means that to *switch* the transistor 'on', a small voltage of about 0.6 Volts is applied to the base connector wire. When the voltage reaches 0.7 Volts, the transistor is fully 'switched on'. In this condition, the transistor is said to be fully '*saturated*'.

The BC 108 is a common general-purpose transistor. The diagram below shows the position of the legs when viewed from underneath the case.

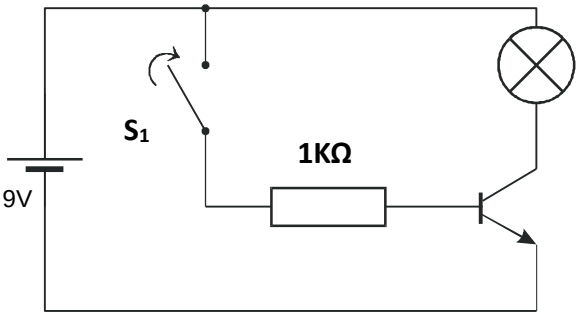


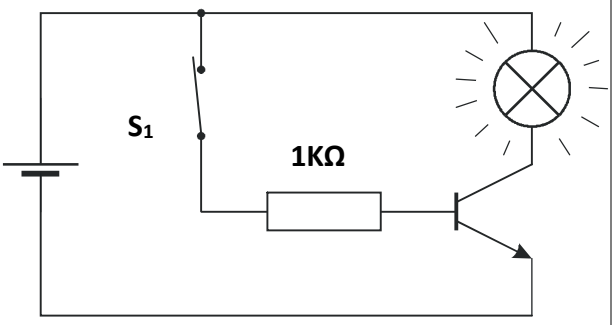



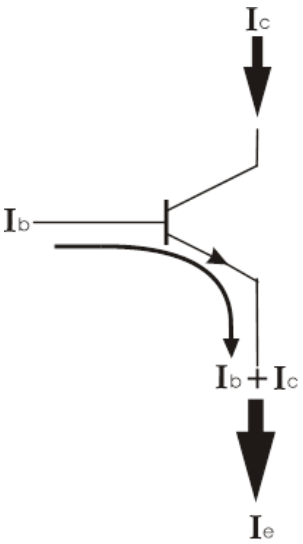

## *How does the transistor work?*

Consider the circuit shown opposite. When the switch $S_1$ is open, no current can flow in any part of the circuit. This may seem strange since a 'complete' circuit appears to be made from the voltage source, through the bulb, the transistor and back to the voltage source. But, as no voltage is being applied to the base of the transistor, it is acting as a barrier to electric current.




When switch $S_1$ is closed, a very small voltage is applied to the base of the transistor. When this happens, the transistor allows current to flow through it and the bulb will light; the transistor is said to 'switch on'.

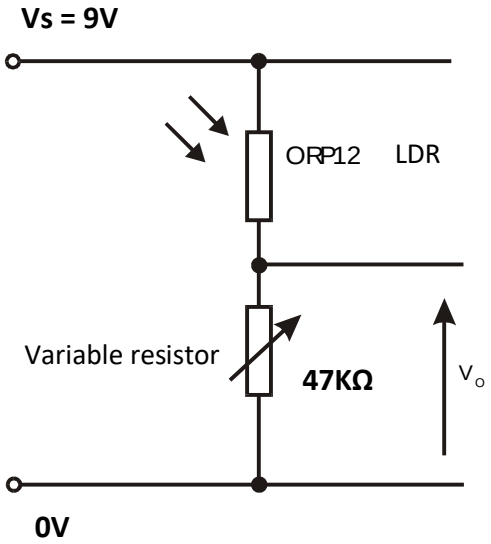



Bipolar transistors amplify current. A small current flowing through the base of a transistor causes a much larger current to flow from the collector to the emitter as shown.

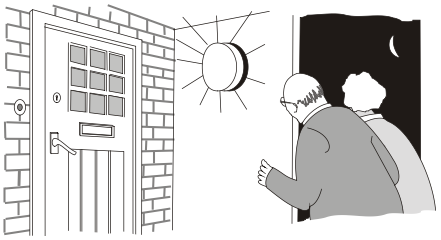
## *Sensitivity*

With an analogue sensor it is normally desirable to adjust the sensitivity of the circuit. Rather than using a fixed resistor we can replace it with a *variable resistor* or *potentiometer*.




## *Voltage divider circuits*

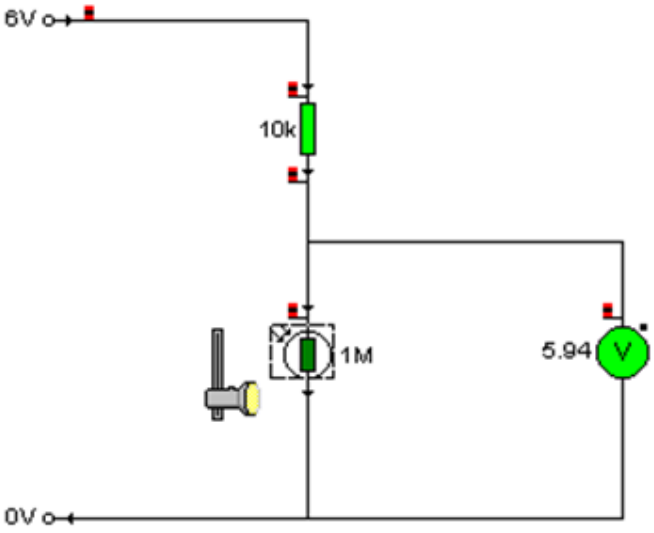
The picture shows a typical situation where a light sensor circuit could be useful. To save money and inconvenience, the residents want the outside light to come on when it gets dark. They also want to be able to adjust the sensitivity from Summer to Winter nights.



**18.** Build the circuit shown below using Yenka. The variable resistor is rated at 10 kΩ.

a) The circuit below senses darkness. As the light level drops across the LDR, its resistance increases. This means the voltage across the LDR will also increase. Prove this actually happens.

b) Alter the circuit so that it senses light. The voltmeter should remain in the same place.

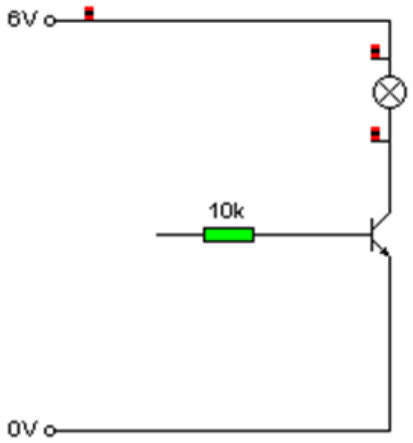


**19.** You have already learnt about explaining the 'system' operation in unit 2. In your jotter, draw both circuit alterations in task 18, with the correct circuit symbols and explain the operation of the circuit in regards to the altering of the light and darkness sensor. Record the voltage across the components.

**20.** The circuit opposite shows a transistor. When it becomes 'saturated', the bulb will light.

a) Build the circuit shown using Yenka.

b) Add a 10 kΩ resistor and an LDR to add a darkness sensor to the circuit. This will light the bulb when the circuit senses darkness. In your jotter, draw both circuit light alterations, with the correct circuit symbols and explain the operation of the circuit as a 'system' of input/process/output.



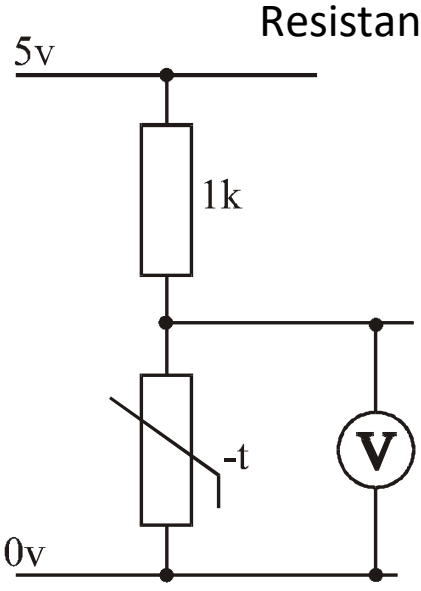

**21.** An NTC (Negative Temperature Coefficient) thermistor is used in a voltage divider circuit as shown below. Using information from the graph below, determine the resistance of the thermistor and hence calculate the voltage that would appear across it when it is at a temperature of:

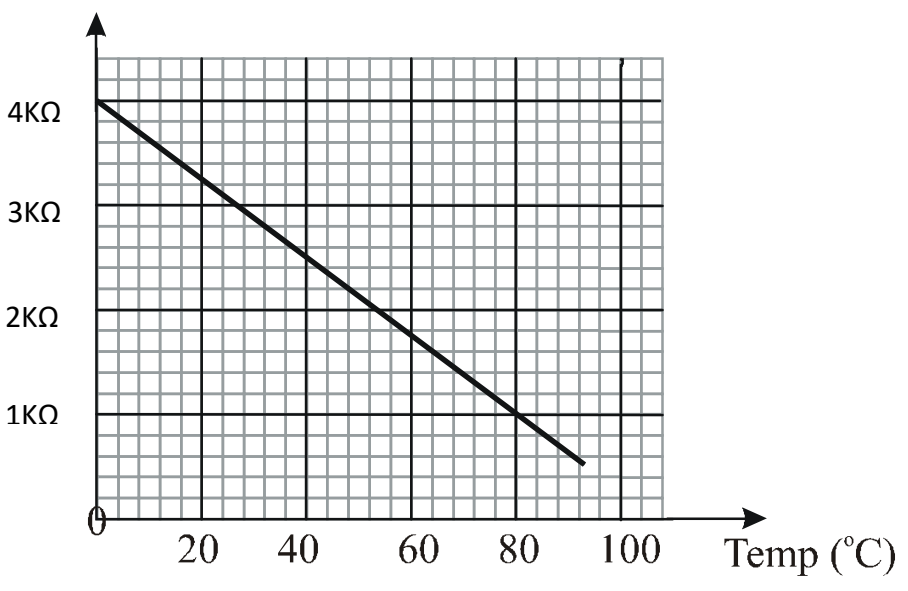
a) 80°C

b) 20°C

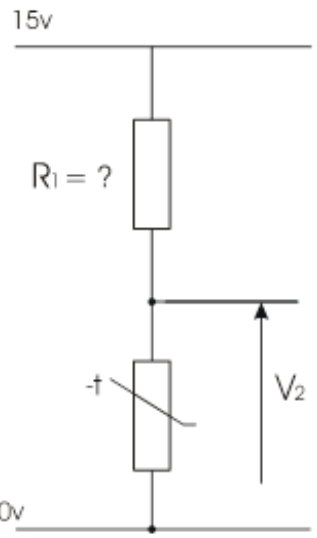
c) What would happen to the voltage across the thermistor in the circuit shown above as the temperature increased?

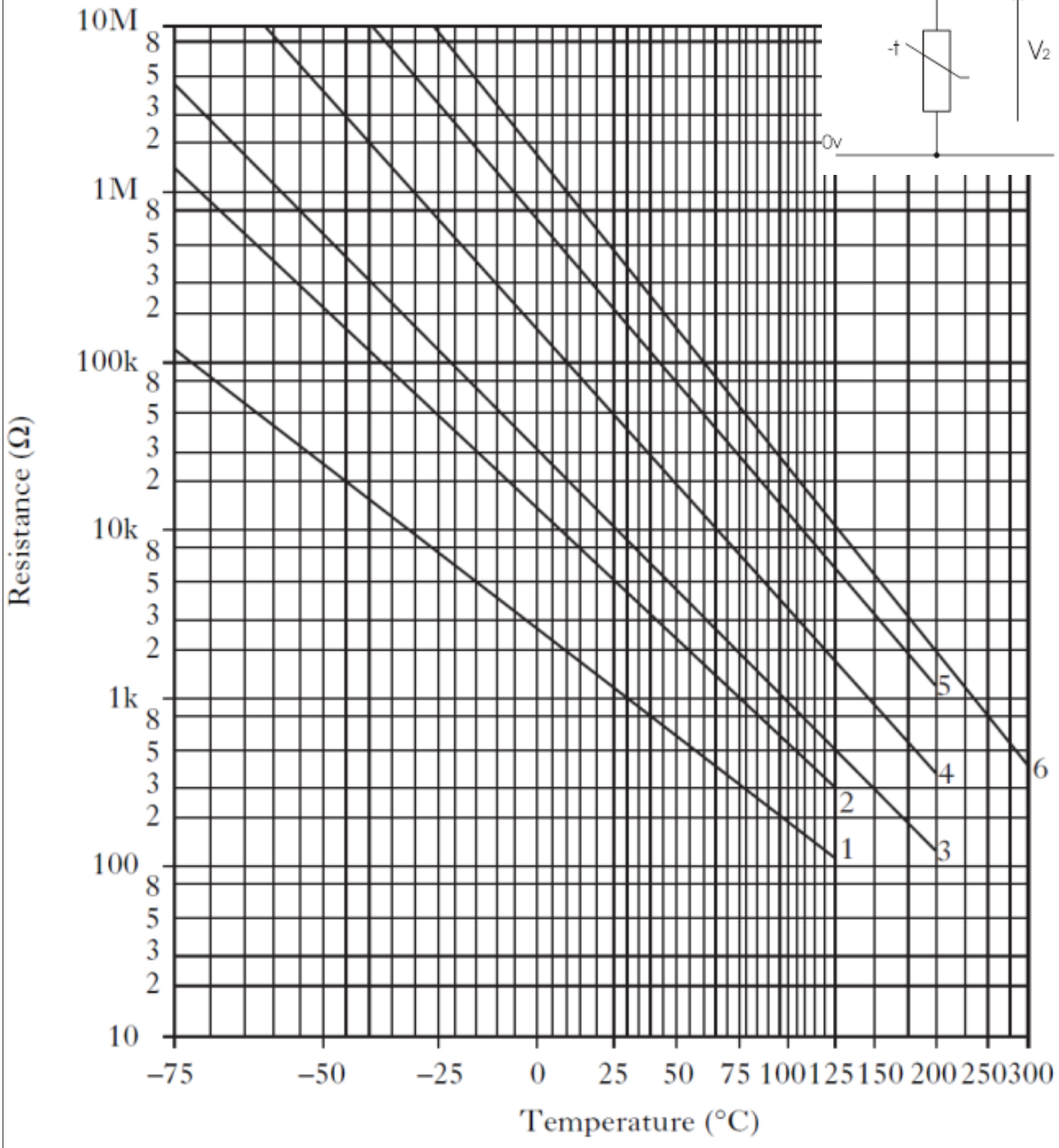
d) What would happen to the voltage across the resistor in the circuit shown above as the temperature increased?



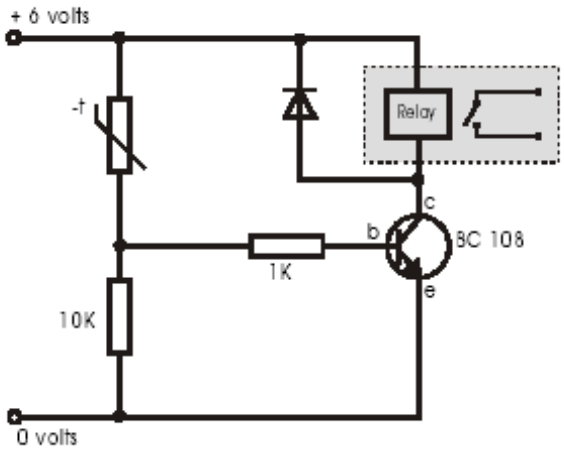
**22.** A thermistor (type 5) is used in a voltage divider circuit as shown below. The characteristics of the thermistor are shown in the graph. If the voltage $V_2$ is to be 4.5 V at 100 °C, determine a suitable value for $R_1$. State whether $V_2$ will increase or decrease as the temperature drops. Explain your answer.



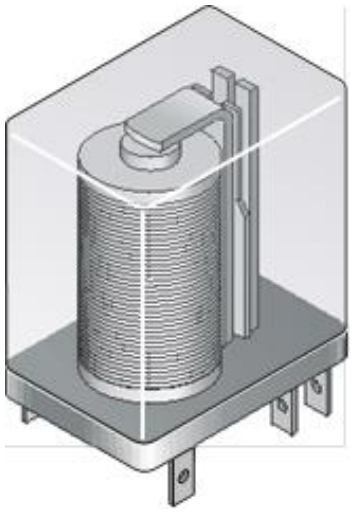



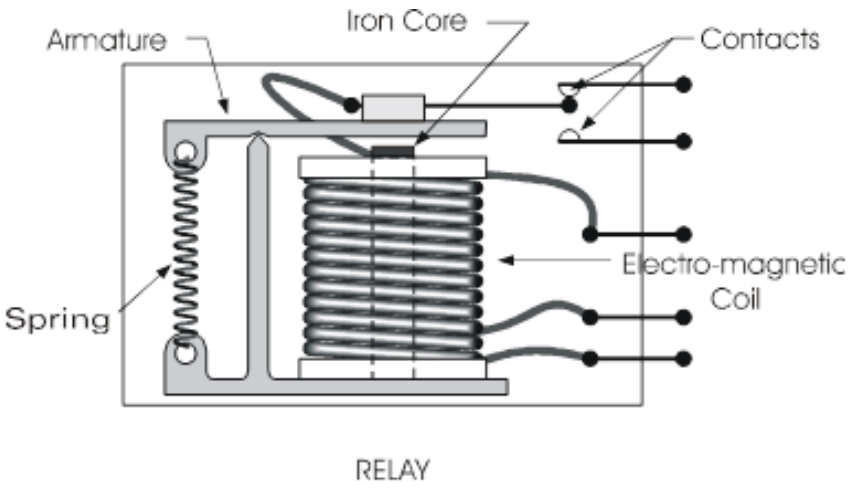
## Components: relays

Although relays are often considered to be output devices, they are really output switches from electric or electronic circuits. These output switches are used as inputs for other circuits. In practice you can hear relays clicking on and off when a car's indicators are used.




## How the relay works



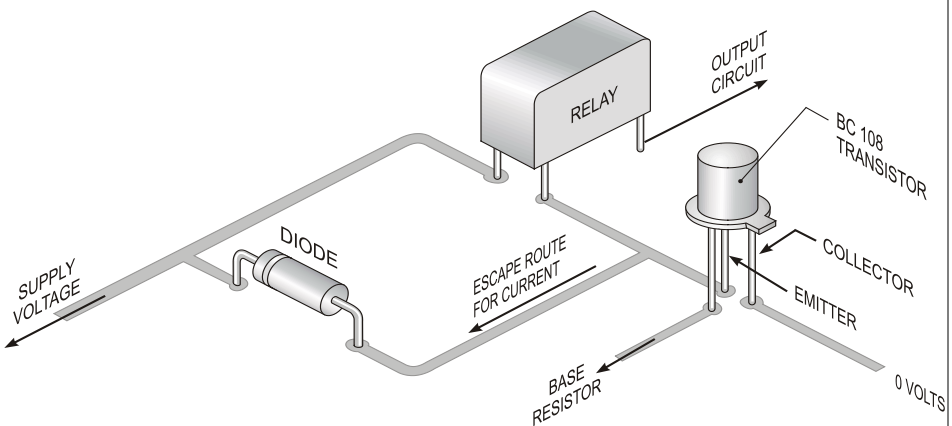



RELAY

When an electric current flows into the relay coil, the coil becomes an electromagnet. This electromagnet attracts the armature and moves the contacts. This movement provides the switching, just as the contacts in any other switch do.

The relay is a very useful device because it is the vital link between microelectronics and high-energy systems that require substantial amounts of current. The relay is perhaps the most commonly used switch for driving devices that demand large currents.

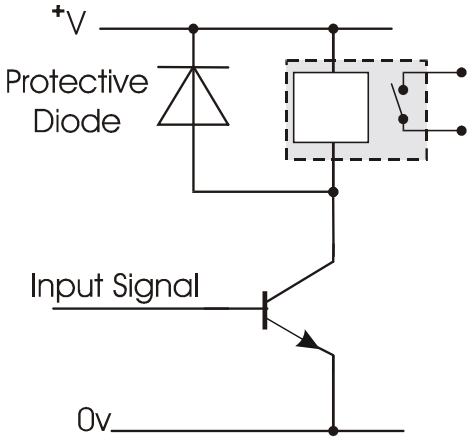
## Relays - protective diodes




As seen earlier, relays have a coil that is energised and de-energised as the relay switches on and off. During this process of switching, the coil can generate a large reverse voltage called **e**lectro **m**otive **f**orce (**emf**). This reverse voltage can cause considerable damage to components, especially transistors. Protection of these components is explained next.

Transistors and other sensitive components can be protected by the inclusion of a diode that provides a path for the current caused by the reverse voltage (emf) to escape.

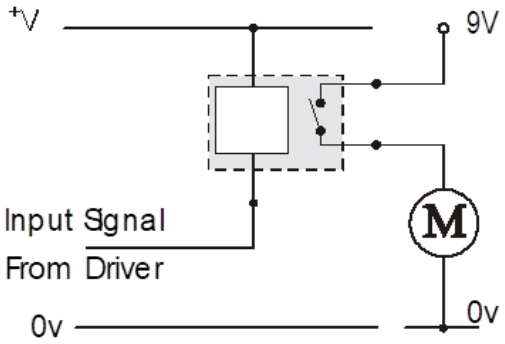
The circuit diagram shows the protective diode.




A solenoid is another output transducer that has a coil inside. Circuits containing a solenoid require a protective diode as well.

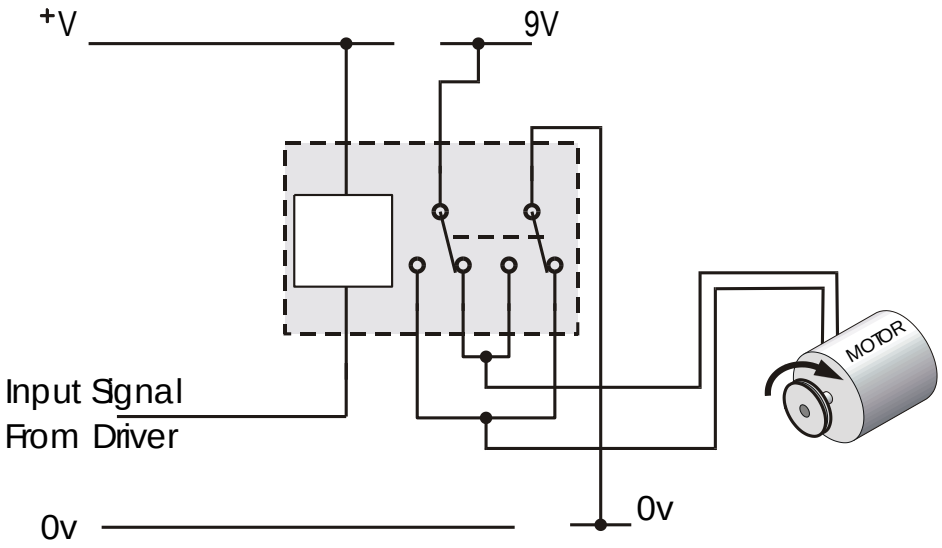
## *Single Pole Single Throw (SPST) relay*

The SPST relay is used for any simple control system. For example, controlling a lamp, turning a heating element on.




## *Double Pole Double Throw (DPDT) relay*

A DPDT relay is ideal for controlling a motor. As can be seen below the motor can run backwards or forwards.



## Components: 555 timers

An integrated circuit is simply an electronic package that contains a number of components on a silicon chip. The 555 Timer IC is versatile and can do many operations. 555 Timers can be used as a monostable device. This means that it is stable in only one state (normally off), and it jumps back to its initial state after a set time.

## Components: Capacitors

Capacitors are electronic components that store electricity for short periods of time within electronic circuits or networks. They are made from 2 metal plates or films separated by an insulator. Capacitors are especially useful in timer circuits with 555 Timer chips. There are two basic types of capacitors normally used in timer circuits; electrolytic and polyester.

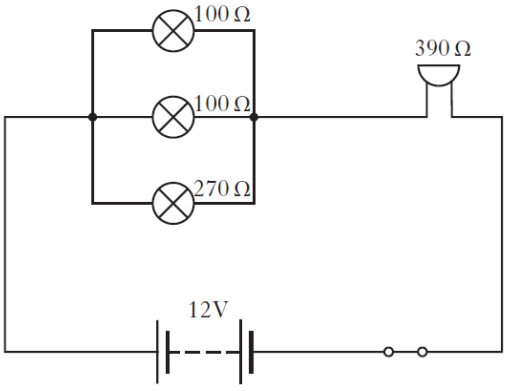
## Components: Stepper motor

Stepper motors are very accurate motors that are commonly used in computer drives, printers, x-y plotters and clocks. Unlike direct current (d.c.) Motors which spin round freely when power is applied, stepper motors require that their power supply is continuously 'pulsed' in four different patterns. It contains magnets that are fixed to the central armature. Four electronic coils are located around the casing. When a current is passed through these coils they generate a magnetic field, which attracts/repels the permanent magnets on coils are then energised in a different pattern to create a different magnetic field and the armature spins another step. For each pulse, the stepper motor moves around one 'step', typically 7.5 degrees (giving 48 steps per revolution).

Stepper motors do have some limitations. First the power consumption is greatest when the stepper motor is stopped (as all coils are still energised). The speed of revolution is also limited to around 100 steps per second, which provides a rotational speed of 2 rev per sec or 120 rev per min.

To make the armature rotate continuously, the four coils inside the stepper motor must be switched on and off in a certain step order. The ULN2803A driver chip on the output driver module provides the method of interfacing these four coils.

**23.** A circuit used to warn when a vehicle is reversing is shown.



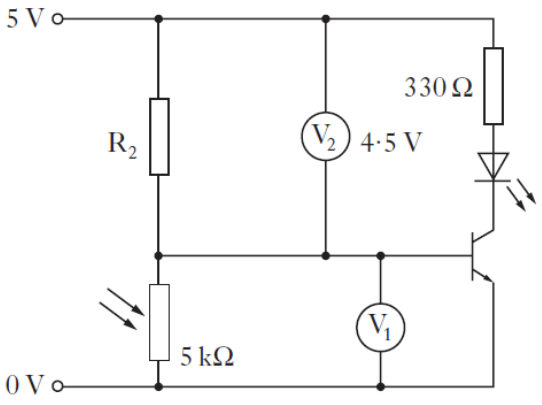

a) Calculate:
   I. The equivalent resistance of the three lamps;
   II. The total circuit resistance
   III. The total circuit current
   IV. The power used by the circuit.

b) Indicate with a cross (X), where an ammeter would be placed to measure the current through the **270Ω** lamp.

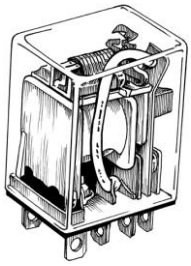
**24.** An electronic circuit for a garden night light is shown. For the conditions shown in the diagram,

Calculate:

a) The voltage shown on $V_1$;
b) The resistance $R_2$;
c) State if the LED is on or off, and explain why this is the case.
*d)* Explain the effect an increasing light level will have on the resistance of the LDR and the voltage shown on V1 (consider system knowledge). 'As the light level increases'….



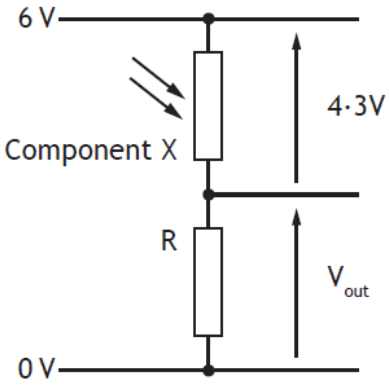

**25.** Relays, like the one shown in the diagram to the right, are often used in electronic control circuits. Describe the general purpose of a relay.



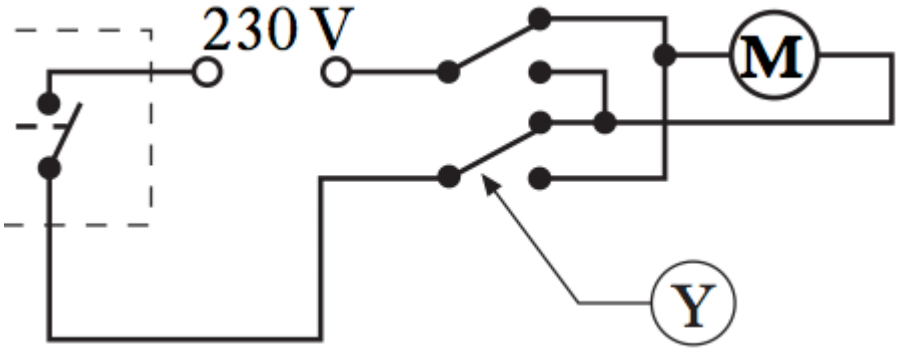
**26.** Using the theory notes but alternative research sources may be used, explain the following:

- What a stepper motor is.
- Where is it used.
- How it operates.

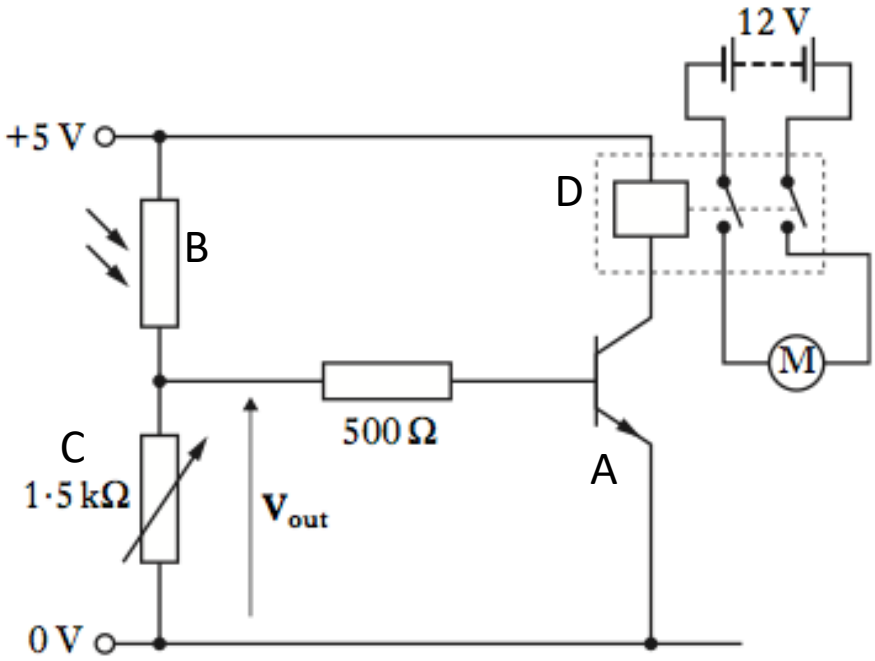
**27.** A weather monitoring station is used to collect data. Light levels are measured using the sensing sub-system shown in the diagram. State the full name of Component X and say what it does.




**28.** The output stage of a control circuit is shown. The DPDT switch **Y** is being used to control the motor. Describe what happens to the motor as **Y** switches positions.



**29.** Part of a prototype electronic circuit that operates an automatic window blind is shown below.




a) State the name of the following components:

***A*** _____ ***B*** _____

***C*** _____ ***D*** _____

b) Describe the function of the following components

A –

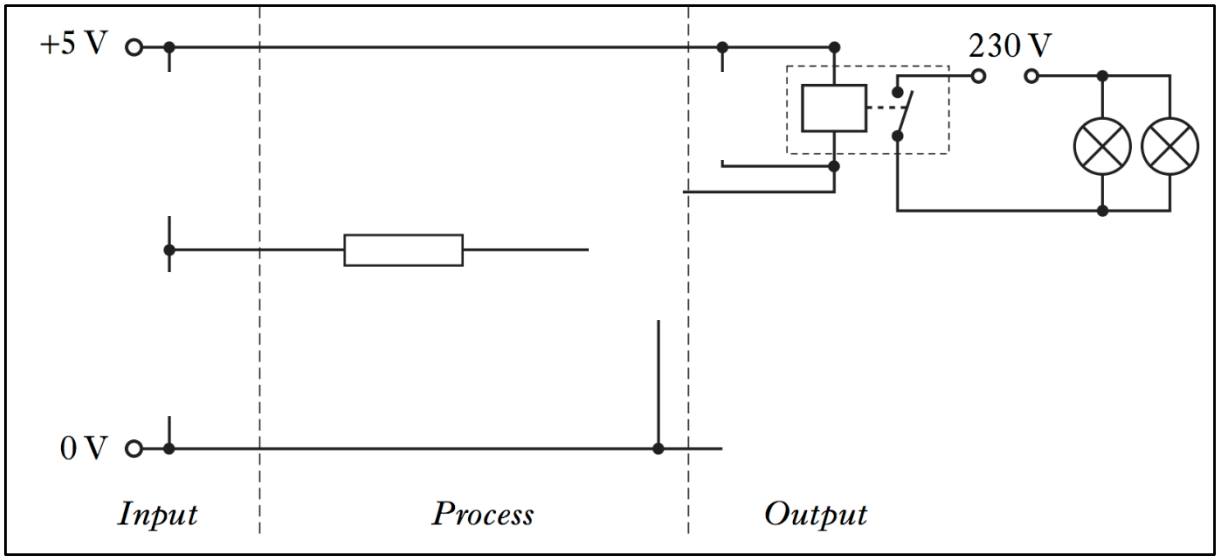
D –

c) Components B and C are connected in series to form a voltage divider. State the function of the voltage divider in this circuit.

**30.** A sports field requires floodlights that will come on automatically when it starts to get dark. An incomplete circuit is shown below.

a) Complete the circuit by drawing the following components in the appropriate places:

I. An LDR and variable resistor in series to form a dark sensor.
II. A transistor.
III. A diode to protect the transistor from back voltage produced by the relay.




b) Describe how the circuit works

**31.** Edinburgh zoo is setting up a new enclosure for their Pandas. A system is required that will control a gate to allow the animals into the outdoor area. The gate should open when it is daylight or if a keeper activates a sensor. It should close when the keeper activates a separate sensor. Some parts of the system will be electronically controlled and other parts will operate using pneumatics.

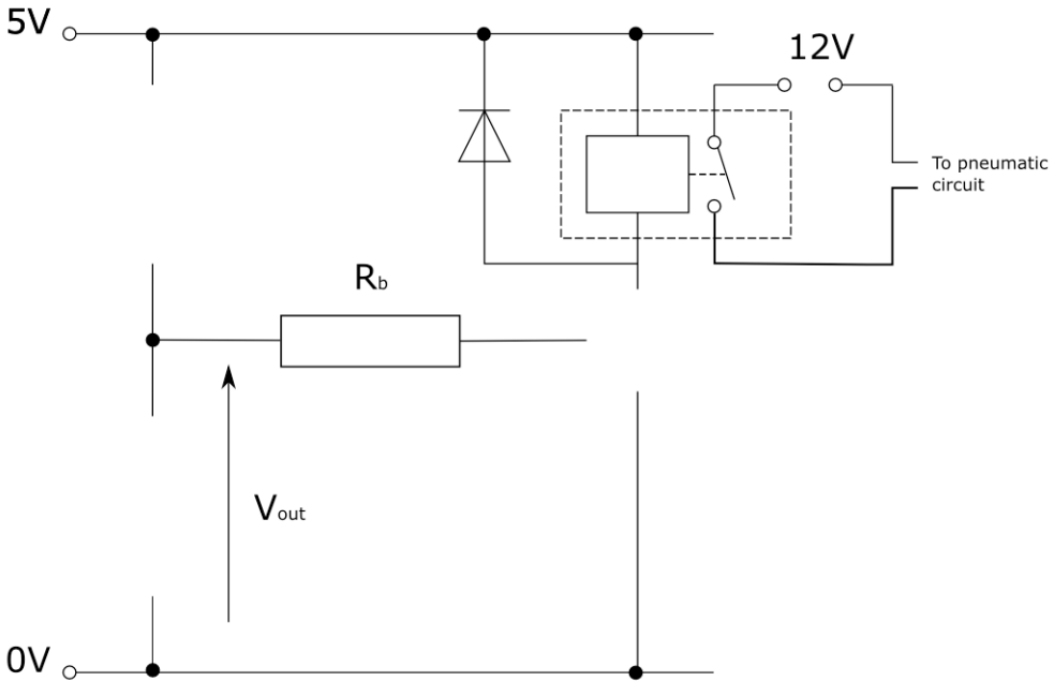


In your jotter:

a) Write a specification for the system highlighting the main inputs, process and outputs.

b) Complete a systems diagram for the enclosure door.

c) An electronic circuit is required that will switch on a relay when it is light. The relay will actuate a solenoid valve in a pneumatic circuit that in turn will open the door. Complete the circuit diagram below.

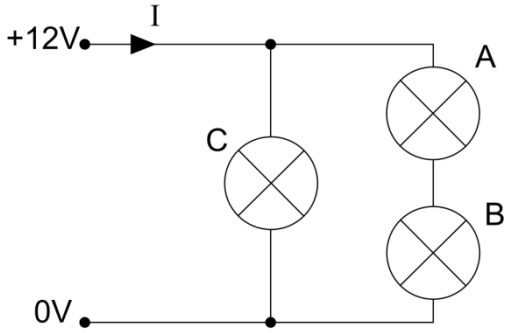



d) Simulate this circuit using Yenka. Attach a print screen in your jotter.

e) Test the circuit and describe how it performed.

f) Compare the circuit's performance against the initial specification, and suggest how it could be adapted or changed to work better.

**32.** Part of a lighting circuit in an animal enclosure is shown below. Lamps A and B each have a resistance of 30 Ω. Lamp C has a resistance of 70 Ω. Calculate:




a) The combined resistance of bulbs A and B.

b) The combined resistance of the circuit.

c) The current (I) from the 12 V supply.

**33.** A local company specializing in energy efficiency is looking to prototype a combined heating/cooling system for use with conservatories. When the conservatory is too warm the system should switch on a fan motor to extract hot air and heat the rest of the house. In cold weather the extractor fan motor will reverse and take hot air from the house to warm the conservatory.



**Problem Specification**

The system must:

- make a motor spin **forwards** when the temperature is **hot**
- make a motor spin **backwards** when the temperature is **cold**
- It must be possible to adjust the sensitivity of the sensor.

a) Apply your knowledge to design a possible solution to the problem brief.

b) Use a computer simulation package to test your circuit design. Attach a print screen in your jotter.

c) Build and test the circuit design using a breadboard and discrete components. Attach a photo.

d) Describe the tests you carried out and how the circuit actually performed.

e) Compare the circuit's performance to the specification. Draw meaningful conclusions and where appropriate suggest possible improvements or further work.